SQCD-1 · Settlement Quality Compute Data

The record a delivered compute hour settles on.

One record covers one delivery claim over one window.

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0 · Status and adoption

SQCD-1 is a record standard. It is adopted by reference: two parties name it in their agreement and it governs the form of the record that settles delivery between them.

No regulator administers it. No body approves a filing under it. Every consequence in this standard is evidentiary — it changes what a record proves, not what a party owes. What a party owes is in the contract.

The words MUST, MUST NOT, SHOULD and MAY carry their ordinary drafting force. A requirement marked MUST is a condition of conformance.

Drafting status. SQCD-1 does not set a value for any term that allocates loss between the parties. A term whose value decides who pays when the record is silent is a commercial term; a value chosen here would apply uniformly to every contract processed under this standard, whatever any of them say, and would sit where neither party's counsel reads it.

Where a party adopts SQCD-1 with modifications, it MUST declare each modification in its Filing Plan. A modification that is not declared makes the filing non-conforming.

0.1 · Adopting it by reference. A contract incorporates this standard by naming the version and the digest of the text:

> *This Agreement incorporates SQCD-1, Version 1.1, identified by the digest published with it, as issued by PSYCHE, and no later version.*

A reference that names no version names nothing this standard will defend. Conformance under L5 is to a named version permanently, and a reference to "SQCD-1" without one leaves the parties to argue later about which text they agreed to. That is the argument this standard exists to remove, moved up one level.

0.2 · Rights in this text. The text of this standard is published under the **Creative Commons Attribution-NoDerivatives 4.0 International licence**: it may be copied, redistributed and attached to an agreement in full, by anyone, for any purpose, including commercially, provided it is unaltered and attributed.

The **forms and schemas** — Annex 1's Filing Plan form, the record schemas Annex 5 names, and any conformance vectors published with this standard — are separately available under the MIT licence, which allows modification, so that an implementer may lift them into code; the licence text is deposited with this version as `LICENSE.md`.

A standard works only where every party is reading the same text, and a licence permitting modified redistribution under the same name would end that. The parts meant to be built from carry no such restriction.

0.3 · Third-party rights. This standard has been drafted without a patent search, and **PSYCHE makes no representation that implementing it infringes no third party's rights**.

The measurement primitive Annex 6 relies on for three V1 fields is third-party work by named authors [ATC], published as a preprint. **PSYCHE holds no licence to it, has sought none, and has received no statement from its authors about patents.** A Profile placing those three fields at V1 relies on that work, and the party doing so is the party exposed.

Where a reader knows of intellectual property that would be infringed by implementing this standard, saying so to PSYCHE lets the next version name it. **A standard that conceals a known encumbrance is worth less than one that discloses it.**

1 · Scope

SQCD-1 applies to a delivery obligation for compute capacity measured over a recurring window and paid in money.

It covers the record. It does not set availability targets, prices, remedies or credit schedules. Those are in the agreement, and this standard requires that they be identified and fixed before the window opens.

It does not apply to the customer's own workload.

2 · Why a record filed by the parties

Two parties dispute a month. Each holds logs, each set is internally consistent, and neither can compel the other to accept its numbers. Six weeks later a figure is agreed on a call, and nobody writes down how it was reached.

Other markets settle this on a record issued by an interested party. A bill of lading is issued by the carrier, who is party to the contract of carriage and has an evident stake in what it says, and banks release payment against it under letters of credit all the same — and the bank has checked nothing about the cargo. It examines the document's face and disclaims responsibility for its genuineness and accuracy [UCP600, Arts 14(a) and 34]. What makes the document worth paying against sits in transport law rather than banking law: **a false bill costs the carrier more than a true one.**

Statute fixes what the carrier must say and lets it decline the rest. Marks, quantity and weight are stated as the shipper furnishes them and warranted by the shipper; condition is stated as the carrier can see it; and the carrier need state nothing it has no reasonable means of checking, which is why bills read *said to contain* [HVR, Art. III rr. 3, 5]. Once said, the statement stops being the carrier's to revise. It is prima facie evidence between carrier and shipper, and **proof to the contrary becomes inadmissible once the bill reaches a third party acting in good faith** [HVR, Art. III r. 4].

Carriers do sometimes sign clean against a letter of indemnity when they know better. Where that is deliberate the courts have held the misstatement and the indemnity procured for it good for nothing, and the carrier answers to whoever relied on the bill [BJPD]. **The doctrine binds the signer whatever the signer knew**, so it works on a party with an interest.

Three properties do the work, and each of them assumes an interested carrier. Form is agreed beforehand, so the issuer cannot choose afterward what to set down and what to omit; the issuer signs, so a false statement becomes its own attributable admission rather than an argument about what anyone remembers; and the hour of signing is established by someone other than the signer, so neither of the first two can be defeated by a convenient clock.

The reservations carry as much weight as the attestation, and this standard reproduces both. A8.10 places every Grade Class field at the tier its filer can actually back: **V3 is *apparent***

order and condition — read through an interface the provider operates — and **V4 is *said to contain***, carried by nothing but the signer's name under A8.11.

Current practice in this market runs the other way, and says so plainly. A practitioner guide to cloud service agreements describes the credit process as one in which the customer supplies proof of the outage and **"the CSP reviews the claims and makes a final, unilateral judgment on service credits"** [CSCC, p. 20]. The party that owes the money decides the claim. F1 and F2 exist to separate those two acts.

This standard requires a record with those three properties, filed by the parties themselves and checkable without either party's cooperation.

3 · Defined terms

Window. The period over which one delivery obligation is evaluated and a Period Figure is produced.

Measured Subject. The unit under measurement, at the grain the contract names.

Grade Class. A named set of specification values under A8. Two compute hours are of the same kind when they belong to the same Grade Class.

Layer. One of the eight strata under B7 at which delivery can fail.

Period Figure. The single quantity produced for the Window by applying the declared function to the filed samples.

Eligible intervals. The intervals of a Window less those excluded under A4 and those removed from numerator and denominator under C5.2 or Annex 2.5. Counted before the C5.2 and Annex 2.5 removals, they are the intervals **eligible before C5**.

Filing Plan. The instrument under Section G by which a party declares, before its first Window, how it will produce everything this standard requires of it.

Filing Party. A party that files measurements. Both the provider and the customer are Filing Parties.

Unresolved Fraction. The share of a Window's intervals eligible before C5 that no Filing Party observed.

Unilateral Fraction. The share of a Window's intervals eligible before C5 carried by exactly one Filing Party's observation.

Divergence. The difference between two Filing Parties' Period Figures for the same obligation and Window.

Delivery Claim. The object under F1. It reports.

Counterparty Decision. The object under F2. It decides.

4 · Conformance

A record is **conforming** when all of the following hold. The conditions below apply together with the other conformance conditions stated in this standard.

4.1 Every field this standard marks MUST is present and populated.

4.2 The Filing Plan in force at the opening of the Window is identified by digest, and the filing is consistent with it.

4.3 The Period Figure recomputes from the filed samples under the function declared in A2, by a party holding no account with either side and using nothing but the filed record.

4.4 The Unresolved Fraction is stated and does not exceed the ceiling declared under A6. **The Unilateral Fraction is stated.** No ceiling applies to it.

4.5 Every filing carries a signature under D4 and an inclusion proof under E2.

4.6 The completeness test. Give the record to someone who was absent and holds no account on either side. If the Period Figure, the shortfall and the amount owed all come out, and each traces to a named field, the record is complete; where a Divergence is carried to decision under Annex 2.7, each candidate figure and the amount it would owe come out, each with its basis. If any step wants a login, a phone call or a person's memory, what you have is a summary of an argument. Where a value is withheld under K4, the test runs on the record the reader receives under an access grant, with the value in it; a record carrying only the value's digest fails the test.

4.7 A requirement marked SHOULD states what this standard expects and what a reader may assume absent a statement otherwise. **A filing that departs from a SHOULD remains conforming and MUST declare the departure under G8.** A permission marked MAY creates no obligation either way, and its exercise needs no declaration under G8.

Completeness puts the figure, or each candidate figure, and its basis in front of the party with authority to accept it.

Part I — The obligation

A · Contract Standard

Evidence takes its meaning from the standard in force, so the standard travels with the record.

A1 · Obligation reference. The clause of the agreement this record answers to.

A2 · Metric definition. The quantity, its unit, and the function that turns samples into a period figure. Two parties averaging differently will produce different months from identical samples.

The function MUST be one of those in Annex 2, identified by its number. A party that requires a function not in Annex 2 MUST state it in full in its Filing Plan, with a worked example over at least one hundred intervals.

A3 · Measurement window. Start, end, offset, and the treatment of samples straddling either edge.

A4 · Exclusions. Conditions that do not count against the obligation, each with the evidence required to invoke it. An exclusion invoked after the window closes is an argument.

A5 · Tolerance and threshold. The shortfall that contributes nothing, and the level at which the obligation is missed.

A6 · Materiality and the unresolved ceiling. Two numbers, both fixed before the window.

The **materiality threshold** is the divergence at or below which two Filing Parties' figures are treated as one.

The **unresolved ceiling** is the Unresolved Fraction above which the Period Figure is not conforming, whatever its value.

The Profile MUST fix both: the materiality threshold by selecting a form from Annex 2.2 and supplying its parameters, the unresolved ceiling by selecting a form from Annex 2.3 and supplying its parameters. **SQCD-1 sets no default for either.** A Profile that omits either cannot be armed. Under MT-01, MT-02, MT-03 or MT-04 the Profile also names, under the A10 digest, the Filing Party whose Period Figure is the reference for an MT-01 fraction and the figure that carries forward under H3; a Profile selecting one of those forms without naming it cannot be armed. Under MT-05 the two figures are equal wherever H3 applies.

A7 · Remedy schedule reference. The conversion from shortfall to money, held where the evaluation cannot read it.

A8 · Grade Class. The kind of compute hour the obligation is for.

A Grade Class is a named tuple over the fields below. Every field carries a value or the token `not-material`. **A blank is not "unimportant"; a blank means the Grade Class is incomplete**, and an incomplete Grade Class MUST NOT be armed.

A8.1 · Accelerator

Field	Expression	Unit
`model`	manufacturer's printed designation	string, as printed
`form_factor`	SXM / PCIe / NVL / other	enum
`memory_capacity`	per accelerator	GB
`memory_type`	HBM3 / HBM3e / other	enum
`memory_bandwidth`	per accelerator	TB/s
`die_topology`	single- or multi-die, with partition	structure string, e.g. `2-die, 74/74 SM`

A8.2 · Intra-node interconnect

Field	Expression	Unit
`link_generation`	NVLink generation or equivalent	enum
`link_bandwidth_per_gpu`	bidirectional, per accelerator	GB/s
`coherent_domain_size`	accelerators in one coherent domain	count

A8.3 · Inter-node fabric

Field	Expression	Unit
`fabric_type`	IB (generation) / RoCE / other	enum
`fabric_bandwidth_per_gpu`	per accelerator	Gb/s
`allocation_contiguity`	`contiguous` / `best-effort` / `unspecified`	enum

A8.4 · Host

Field	Expression	Unit
`host_threads_per_gpu`	processor threads ÷ accelerators	count
`host_memory_per_gpu`	GB ÷ accelerators	GB
`local_nvme_per_gpu`	capacity and sequential throughput floor	TB, GB/s

A8.5 · Attached storage

Field	Expression	Unit
`storage_throughput_floor`	per node or per accelerator; which one MUST be stated	GB/s
`storage_iops_floor`	as above	IOPS

A8.6 · External network

Field	Expression	Unit
`egress_floor` / `ingress_floor`	per node	Gb/s

A8.7 · Locality

Field	Expression	Unit
`region` / `availability_zone`	provider's naming	string
`power_region`	grid balancing area	string
`placement_granularity`	`rack` / `hall` / `site` / `region`	enum

A8.8 · Tenancy

Field	Expression	Unit
`tenancy`	`dedicated` / `shared`	enum
`preemptibility`	`reserved` / `preemptible`	enum
`isolation_attested`	confidential-computing mode enabled and the attestation verified	enum

A8.9 · Software floor

Field	Expression	Unit
`driver_floor` / `firmware_floor` / `runtime_floor`	version at or above which the obligation is measured	version

A rate quoted without a Grade Class prices an unnamed good. Two providers printing the same number are selling different things when one of them ships a downlevel driver or a fabric with half the bandwidth, and both are telling the truth.

A8.10 · Verifiability. Every field carries a verifiability tier. The tier is a property of the field, not of the parties, and it decides what the Profile must require in evidence.

Tier	Definition
V1 · Physically measurable	The customer establishes the value by measurement, on the allocation it holds, **without the provider's cooperation and without a vendor key**. Misreporting requires defeating a physical measurement.
V2 · Attestable	A hardware root of trust signs the value and the customer verifies the signature against a manufacturer's certificate chain. Requires vendor attestation support and provider enablement. The customer verifies an endorsement, not a physical property.
V3 · Observable, provider-mediated	The customer reads the value through an interface the provider operates — driver, hypervisor, control plane, cgroup. **Detects error and misconfiguration; does not survive an adversarial provider.**
V4 · Declared	No procedure in Annex 7 reaches the value and no attestation carries it. The field is established by a signed statement under A8.11.

Memory capacity as the driver reports it is V3: a number the provider's process tells you. **Memory bandwidth as measured is V1: a physical quantity the provider cannot fake without supplying it.**

Field	Tier
`memory_bandwidth` · `link_bandwidth_per_gpu` · `coherent_domain_size` · `fabric_bandwidth_per_gpu` · `local_nvme_per_gpu` throughput · storage floors · network floors	**V1**
die identity · `die_topology` class · coarse geographic position	**V1**, on an unprivileged measurement primitive. See Annex 6.
`model` · `isolation_attested`	**V2**, where the platform's confidential-computing mode is enabled; **V3** otherwise. The device's signed attestation report carries a hardware-model claim, and the report's firmware measurements are compared against manufacturer-published reference values. See Annex 6.
`form_factor` · `memory_capacity` · `memory_type` · `link_generation` · `fabric_type` · `host_*` · `local_nvme_per_gpu` capacity · `region` / `availability_zone` / `power_region` names · `placement_granularity` at `region`	**V3**
`driver_floor` / `firmware_floor` / `runtime_floor`	**V2** where the attestation report's measurements are checked against the manufacturer's published reference values; **V3** otherwise
`allocation_contiguity` · **`tenancy`** · **`preemptibility`** · rack-level position · `placement_granularity` at `rack`, `hall` or `site`	**V4**

A8.10.1 · What the tiers measure, and what they run together. A tier answers three questions at once: who produces the evidence, whether a manufacturer vouches for it, and whether anyone independent appraises it. Those come apart. A provider that cryptographically signs its own control-plane report is producing better evidence than one that does not, and this ladder scores both as V3 because neither carries a manufacturer's endorsement. **A Profile that wants the distinction MUST state it as a required evidence class rather than expect the tier to carry it.**

The difference that survives every decomposition is the one between V1 and everything below it. **V1 asks the customer to trust a physical property. V2 asks it to trust a manufacturer. V3 asks it to trust the counterparty. V4 asks it to trust a signature and a date.**

A8.11 · The V4 rule. A Grade Class field at tier V4 **MUST** carry a required evidence class under the Profile.

Where a dimension cannot be measured, this standard requires the party that knows to sign a statement of it, before the window opens, under the A10 digest.

Latency gives a distance ordering rather than a rack-level position, and a customer who bought a contiguous domain and received a scattered allocation has no procedure here that establishes it.

`tenancy` requires more care than one word can carry, and this standard separates three claims. Whether the customer holds a die exclusively, across time, remains unestablished by any mechanism known to this standard, and stays V4. Whether the customer's work is cryptographically isolated from everything else on the machine, including the provider, **is attestable today** on parts whose confidential-computing mode is enabled, and this standard carries it as the separate field `isolation_attested` rather than folding it into `tenancy`. And whether another tenant is in fact present has been shown to be detectable from inside a tenancy, without privilege and without the provider's help, by published and peer-reviewed work on co-residency detection [CORES] and on contention across a shared interconnect [GPUBOX]. **Those results are an existence result and not a deployed evidence class**: no procedure in Annex 7 implements them, and none carries 7.0.1's declared conditions and the bounds in 7.1. A Profile MUST treat `tenancy` as V4 until one does.

The carrier is asked to sign early, on a form fixed in advance, to something it cannot later revise. **An exclusivity declaration is the same instrument: fixed before the window, attributable, and — when it turns out to be false — a signed document that is false, rather than an argument about what anyone remembers.**

A9 · Substitution and netting. What may be set against what.

A9.1 Surplus on one Measured Subject MUST NOT offset deficit on another, unless the agreement names the aggregate as the Measured Subject under B3.

A9.2 Hours of different Grade Classes MUST NOT be netted or substituted without an adjustment factor selected from Annex 2.6 and fixed before the window.

A9.3 Where the agreement permits aggregation, the record MUST carry both the aggregate figure and the per-subject decomposition. An aggregate that cannot be decomposed can only be accepted or refused.

This is where two contracts that both promise the same percentage stop meaning the same thing. One counts per accelerator; the other counts across a purchase order. Both instruments read correctly, and a customer can lose a rack without the number moving.

A10 · Standard digest. A single commitment over A1 through A9, carried in an event that precedes the Window.

Every other clause in this Section describes a term. This one describes the moment the terms stop moving. Without it, any of the nine can be edited under an unchanged version number, and the edit rewrites the standard for Windows already closed. The record would carry no trace of it, because each filing would still be consistent with the standard as it then read.

Every declaration made before the Window is signed under A10. A10 also makes the bill-of-lading argument in Section 2 available to this standard rather than merely cited by it: form agreed beforehand is form committed beforehand, and a commitment is a digest.

Part II — The record

B · Measurements

Both sides file. Provider telemetry and customer-side records arrive as separate submissions and stay separate.

B1 · Channel. The obligation this series measures.

B2 · Samples at the recorded interval. The series as the instrument produced it, ahead of any summary. A monthly figure that resists decomposition can only be accepted or refused.

B3 · Measured subject. The unit under measurement, at the grain the contract names: node, host, rack, cluster, endpoint. A customer who bought a rack and a clause that measures a host are describing different objects, and both instruments can be read correctly at the same time.

B4 · Measuring party and instrument. The party that took the reading, and what it read from.

B5 · Clock source and stated accuracy. Two logs that disagree about when will disagree about how much.

B6 · Aggregation applied before filing. Any downsampling between the instrument and this record, and the function it used.

B7 · Layer. The stratum at which this series observes, from the following list.

	Layer	Observes
L1	Facility	power, cooling, space
L2	Host	node reachable, firmware, thermal state
L3	Accelerator	present, healthy, at declared clocks
L4	Interconnect	intra-node domain and inter-node fabric
L5	Storage	attached volumes reachable at the declared floor
L6	Network	ingress and egress at the declared floor
L7	Control plane	work can be placed and started
L8	Tenant workload	the customer's own code

B7.1 The obligation MUST name the Layers it covers.

B7.2 An interval is delivered only where **every covered Layer was simultaneously conforming**. A conforming reading at one Layer is not an answer about another.

B7.3 L8 MUST NOT count against the provider.

B7.4 Where more than one covered Layer failed in an interval, the record MUST name each. Attribution then follows A4.

A machine can hold its benchmark and still deliver nothing, because the scheduler never placed the work. Measuring the accelerator and reporting the month is measuring one Layer and reporting every covered Layer.

B8 · Minimum customer-side observation. What a customer must observe on an interval.

B8.1 A customer that files nothing observes nothing, and an obligation measured only by the party that owes it is measured by one party.

B8.2 The agreement MUST name the minimum customer-side series. In the absence of a named minimum it is: reachability of each Measured Subject at an interval no coarser than the provider's, and the customer's own record of work submitted, started and completed.

B8.3 A customer that does not meet B8.2 for a Window may still file. Its filing is recorded, and Section H applies to it as to any other.

B8.4 · Grade verification. Where the obligation is stated in a Grade Class, the minimum also includes a reading of each **V1** field of that class, taken under a named procedure in Annex 7.

B8.5 · Cadence. A V1 reading is taken at commencement, and thereafter at the cadence the Profile fixes. The commencement reading establishes that each V1 field of the Grade Class contracted for was delivered. Later readings establish that those fields were still delivered when the readings were taken.

C · Missing Intervals

Disagreement concentrates where one party has data and the other does not. An interval with no observation enters the record as a signed statement.

The canonical form of a gap is the **BoundaryRecord** of the reference implementation; the fields below name its parts.

C1 · Interval. Start and end of the period without data. A gap that is instantaneous rather than extended is recorded as a point.

C2 · Cause. A code, not free text, drawn from the controlled vocabulary: `instrument-down`, `instrument-failure` (a measurement run that produced no reading, Annex 7.0.3), `retention-expired`, `collector-restarted`, `never-instrumented`, `scope-excluded`. A record carrying a cause outside the vocabulary is not conforming.

C3 · Attribution. The party whose evidence is absent, and the subject the gap is against.

C4 · Scope, materiality and signature. The scope the gap falls in — channel, external receipt, spot check, or global — its materiality, and a signature over C1 through C4. **A gap counts as declared only when signed, and an unsigned gap of a party required to observe is an omission under C5.4.**

C5 · Value assigned. What number an interval eligible before C5 carries into the Period Figure.

C5.1 Where one Filing Party observed the interval and the other did not, the record carries the observation that exists, marked **unilateral**, and the interval counts toward the Unilateral Fraction. Whether a unilateral observation determines the interval follows the Profile's Annex 2.5 selection for that obligation, until a measurement the first paragraph of J2 admits for the pass covers the interval. **SQCD-1 sets no default.**

C5.2 Where no Filing Party observed the interval, and every Filing Party whose Filing Plan required the observation declared the gap under C1 through C4, the interval is **Unresolved**. An Unresolved interval MUST be excluded from both the numerator and the denominator of the Period Figure, and MUST be counted in the Unresolved Fraction.

C5.3 Where the Unresolved Fraction exceeds the ceiling in A6, the Period Figure is not conforming, and no amount owed rests on it until the parties agree a figure by a route outside this standard.

C5.4 An interval that a Filing Party was required by its own Filing Plan to observe, and neither observed nor declared under C1 through C4, is an omission of that party: it is attributed to that party under C3, and J1 applies. C5.4 is read before C5.2: a declared gap is an account of what was not observed, an undeclared one is an omission. Where more than one Filing Party was so required and did not declare, each bears its own fault. What the interval then carries is assigned under C5.5.

C5.5 · Determination. Every interval falls in exactly one row.

Observed by	Declared under C1 through C4	Value assigned	Fault under J1
both parties	—	the observations; Section H where they diverge	none
one party	—	C5.1: the observation that exists, marked unilateral, per the Profile's Annex 2.5 selection	the non-observing party, where its Filing Plan required the observation and it did not declare
no party	every required party	C5.2: Unresolved; excluded from numerator and denominator, counted in the Unresolved Fraction	none
no party	some required party did not	the Profile's Annex 2.5 selection, read under 2.5.2	each required party that did not declare

Counting an interval in the Unresolved Fraction states that no observation reached the record; a fault recorded beside that count says why. The count never reclassifies the fault, and the fault never supplies a reading. A measurement the first paragraph of J2 admits for a pass moves the interval, in that pass, to the row its observations then fill, and the fault recorded under J1 at W+2 stays with it; a measurement J2 archives moves nothing.

D · Provenance

Every filing is attributable to one party, and the record names the class of evidence it came from.

D1 · Filing party and signing key. One key per party. Nothing is filed on another party's behalf.

D2 · Source class. Provider-side, customer-side, hardware-attested, third-party-operated.

D3 · Independence. Whether two required sources share a trust root. Two readings from one root are one reading.

D4 · Signature over the filed set, algorithm named. Alteration after filing becomes detectable rather than deniable.

E · Temporal Binding

Everything above depends on when a record was made, and an interested party can claim any hour it likes.

E1 · Signing time as claimed. It is necessary, and worth nothing by itself.

E2 · Inclusion proof against a published log head. A record inside a head published at time T existed before T, whatever its own clock asserted.

F · Claim and Decision

The two objects have different signers. One reports, the other decides.

F1 · Delivery claim. Window, standard, subject, the pass whose Period Figure it rests on under I.1.4.1, and the value at risk. The quantity it claims is the one carried forward for that pass under H3, or under H4 and the Profile's Annex 2.7 selection where the Divergence is material, recomputed from the filed samples under A2, and its evidence set is the evidence receipts that name it. Where the selected Annex 2.7 path has not yet decided the quantity, the Claim names the pass's candidate figures, each with its basis, as 4.6 and Annex 3.6 allow, and leaves the choice among them to that path. It carries no verdict and no settlement status.

F2 · Counterparty decision. Accept, reject, dispute, or abstain. Signed, attributable, and bound to the object it turns on. This field alone produces `accepted`, once the accepts filed reach the quorum of the armed settlement policy, and a reject or a dispute filed here is terminal at once; an abstention, or accepts below quorum, leave the Case to I.1.1; `expired` comes from the closing of the review window or the timeout at the Case deadline, as I.1.1 derives it, and from nothing else.

F3 · Authority reference. The grant F2 was signed under, and the limits of that grant. A signature given outside its grant binds no one, and the party that discovers this discovers it late.

Part III — Filing, divergence, and finality

G · Filing Plan

How everything above is produced is declared before the first window, by the party that will produce it.

A Filing Party MUST file a Filing Plan before its first Window. A filing made under no Plan is not conforming.

G1 · Filing party and role. Provider or customer, and the obligations this Plan covers.

G2 · Grade Classes. Each Grade Class under A8 this Plan produces records for, with its values.

G3 · Instruments and Layers. For each covered Layer under B7: the instrument, what it reads from, the sample interval, the clock source, and its stated accuracy.

G4 · Collection, validation and aggregation. How readings travel from the instrument to the filing, including every validation, estimation or editing step and the function applied at each.

G5 · Period function. The Annex 2 number, or the function in full where Annex 2 does not contain it.

G6 · Minimum observation. For a customer, how B8 is met. For a provider, the coverage its instruments achieve across the Measured Subjects.

G7 · Audit and test schedule. When instruments are tested, by whom, and against what.

G8 · Declared modifications. Every departure from this standard, section by section. An undeclared departure makes every filing under the Plan non-conforming for as long as the departure stood.

G9 · Plan digest and effective window. A digest over G1 through G8, and the first Window it governs.

G10 · Change control. A modified Plan takes effect at the opening of a Window and never inside one. A party may file under its existing Plan while a modification is pending, so long as it remains conforming under the version in force. A Plan digest that changes without a new effective Window rewrites months already closed. Under a Profile that selects MT-02, every Plan in force for a Window samples at the A3 sample interval; a modification that would set another interval is not conforming and does not take effect, and the party files under the Plan in force.

H · Divergence

Electricity must balance, so a discrepancy surfaces on its own. Compute balances nothing. What is not filed is not noticed.

H1 · Duty to file regardless of agreement. Each Filing Party MUST file for every Window, whether or not it believes there is anything to dispute. A record produced only when someone complains measures complaints.

H2 · Divergence computed. For each obligation and Window, the difference between the Filing Parties' Period Figures, computed by either party from the filed record.

H3 · Concordance. Where Divergence is at or below the materiality threshold in A6, compared as Annex 2.2.1 states for the selected form, the record states concordance and names the figure that carries forward: under MT-01 to MT-04 it is the Period Figure of the Filing Party the Profile names under A6; under MT-05 the equal value of the two Period Figures carries forward.

H4 · Divergence Event. Where Divergence exceeds the threshold, compared as Annex 2.2.1 states, a Divergence Event MUST be filed. It carries:

1. both Period Figures;
2. the interval-level decomposition of each;
3. **the first interval at which the two series part**;
4. the Layers and Measured Subjects at which they part;
5. the Unresolved Fraction as each party states it.

How a material Divergence is carried to one quantity is a term of the agreement. The Profile MUST fix it by selecting from Annex 2.7.

H5 · No complaint required. A Divergence Event arises from the arithmetic. Neither party requests it, and neither party can decline it.

Localising a disagreement to its first interval ends most of them. Two parties who cannot agree about a month can usually agree about a minute.

I · Timetable and Finality

A record that never closes is a record that never settles. Two clocks run, because two audiences read.

I.1 The invoice clock. Days are business days after the close of the Window, unless the agreement states otherwise. A review deadline the agreement sets falls after the delivery Claim is filed and before W+30; a Profile that sets one outside that span cannot be armed.

	Event
W+2	Each Filing Party files measurements under B and C.
W+5	Divergence computed under H2. Any Divergence Event filed.
W+7	Delivery Claim filed under F1.
W+12	Counterparty Decision due under F2. Absent a Decision, I.1.1 applies.
W+30	**Correction window closes.** Subject to I.1.2, the correction window opens at the review deadline, and either party may file corrected measurements in it until this day. The record as corrected by this day yields the corrected figure, identified under I.1.4.1 by this day's date. The Claim decided at W+12 and the Case it opened stand as decided; the difference the correction establishes is claimed by the party it favors as a credit Claim under I.1.5 and the agreement's credit terms, or recorded as an amendment both parties sign.
W+45	**Quantity finality.** No measurement may be filed or amended for the Window after this day. A measurement filed later is not part of the record.
W+90	**Money finality.** The Period Figure and the amount owed are final between the parties.

I.1.1 Silence. Where no terminal decision has taken effect under the armed settlement policy by the review deadline, the Case does not become accepted. Under that policy an accept takes effect once the accepts filed reach the policy's quorum, and a reject or a dispute takes effect at once; a Decision counts toward either only where its signature is valid and it was given within a grant under F3, and each of those two tests stands apart from the quorum. An abstention does not by itself end the Case, and accepts below quorum at the review deadline leave the Case to this clause. Silence resolves the Case through the terminal state `expired`, derived from the event that shows the deadline passed, the closing of the review window or the timeout at the Case deadline, and never through implicit acceptance.

What expiry costs the silent party is a term of the agreement. The Profile MUST fix it by selecting from Annex 2.4. **SQCD-1 sets no default.**

A handler that reads silence as consent manufactures the one thing this standard exists to keep unmanufactured. Acceptance is an act by a party holding standing. An authority that was not exercised was not exercised.

I.1.2 A Decision to dispute filed by the review deadline opens the correction window early. It does not move quantity finality, and it does not reopen the disputed Case: the corrected figure enters the record as the W+30 row states.

I.1.3 After money finality the figure moves only where both parties file a signed amendment naming this section.

I.1.4 · Recomputation passes. The timetable above runs one correction window. A Profile MAY state further passes between the Claim and quantity finality, each with its own date, each producing a Period Figure of its own under I.1.4.1.

I.1.4.1 Every Period Figure is identified by its pass, and recomputed under A2 from the measurements filed by the pass's date. A pass is identified by its date, written as I.1 writes it, such as W+30. A Claim names the pass whose Period Figure it rests on. The single pass in I.1 is dated W+5, when H2 computes Divergence on it, and a Claim resting on it names no pass; its corrected figure under the W+30 row is identified by W+30; a further pass is identified by the date the Profile states for it. A later figure leaves an earlier one in the record, and a Claim already decided stands as decided: the difference a later figure establishes is claimed under I.1.5.

I.1.4.2 Quantity finality under I.1 binds the last pass. A pass dated after W+30 recomputes only from the measurements filed by W+30.

Every settlement-quality metering regime this standard was drawn from recomputes on several published dates rather than once — one grid operator runs four passes inside a hundred and five days [MISO], and the regime SQCD-1 takes its name from issues an initial statement and four scheduled recalculations [CAISO]. **The single-pass timetable in I.1 is a choice about compute's operating tempo, not the field's practice**, and a Profile whose obligation collateralises a facility will probably want more than one. A multi-pass regime demands in return that a reader holding a Period Figure can tell which one it is, and I.1.4.1 supplies that.

I.1.5 · Credit Claims. A credit Claim under the W+30 row names the Window, the pass whose figure establishes the difference, and the Claim it adjusts, and is filed only once every earlier credit Claim for the same obligation and adjusted Claim has reached a terminal state. Its amount is the remedy quote of its pass, less the adjusted Claim's amount, which for this clause is the remedy quote of the pass that Claim rests on, less the amounts of the credit Claims for the same obligation and adjusted Claim that ended `accepted` or `expired` under EX-03; each amount counts as positive where it favours the party the remedy schedule pays and as negative where it favours the other. A credit Claim that expired under EX-03 stays `expired`; what is deducted for it is the amount owed under that form, which is deducted because it is owed, once, whether or not it has yet been paid. Credit Claims that ended `rejected` or `withdrawn`, or `expired` under a form other than EX-03, are not deducted. A positive amount is claimed by the party the remedy schedule pays and a negative one by the other party; where 2.1.4 leaves either pass without a figure, the amount follows the agreement's terms identified under A7. Its review deadline falls one review span after its filing: the span from the filing of the delivery Claim to its review deadline under I.1, five business days on I.1's own dates, unless the Profile states another span for credit Claims, which is at least one business day; a Profile stating a shorter one cannot be armed. A credit Claim is not conforming where it is filed before an earlier one is terminal, where its review deadline would fall on or after W+90, or where an earlier credit Claim for the same obligation,

adjusted Claim and pass ended other than `withdrawn`. A credit Claim that ended `withdrawn` keeps its identifier and stays in the record as `withdrawn`, and only that state leaves the pass open to a resubmission; `rejected`, `expired` and every other terminal state close it. A resubmission is identified as Annex 5 states, rests on the same pass's figure, recomputed from the measurements filed by that pass's date, and is bound by the W+90 limit above. I.1.1 applies to a credit Claim at its own review deadline.

I.2 The covenant clock.

	Event
Q+10	**Period Attestation.** A named officer of each Filing Party affirms in writing that the Filing Plan in force was implemented and followed for every Window in the quarter save those it identifies, and identifies every Window in which it was not.
Annual	**Self-audit.** Each Filing Party audits its own instruments and the path from instrument to filing, against its Filing Plan, and files the result. A party may engage an independent qualified firm to conduct it.

I.2.1 A Period Attestation MUST NOT be given for a quarter in which the party failed to file under H1 for more than the unresolved ceiling in A6, measured across the quarter's Windows.

The invoice clock is for the two parties. The covenant clock is for whoever lent against the contract, and the Period Attestation exists because a lender testing coverage every quarter cannot read a monthly invoice thread.

J · Consequences

This Section fixes the consequences of the I.1 filing timetable and how evidence faults are allocated. No fine is available here: what a late filing costs is the benefit of the doubt.

SQCD-1 imposes no penalty and collects nothing. Every consequence below is evidentiary. Where parties want a monetary consequence, it belongs in A7.

J1 · Evidence-fault allocation. A Filing Party that, for an interval its Filing Plan required it to observe, files neither conforming measurements nor a declaration under C1 through C4 by W+2 produces an evidence fault on that interval.

Who bears the interval on which an evidence fault falls is allocated by the Profile, **per obligation**, by selecting one form from Annex 2.5; a Profile selecting more than one form for one obligation cannot be armed. **SQCD-1 sets no default and states no global rule.**

The allocation decides whether withholding pays. If a fault suspends payment, a buyer gains by suppressing its own scheduler records; if a fault admits the claim, a provider gains by suppressing its telemetry. The incentive moves to whichever party first notices the standard was silent.

J2 · Late filing. A Filing Party's first measurement of an interval, filed after W+2 and by W+30, enters the record marked late. It does not displace an observation already filed on time where the two conflict, unless the counterparty accepts it. Subject to that, the interval it covers is valued from the observations in the record, the late measurement included, rather than from the Profile's Annex 2.5 selection; where that selection is EF-07 with its estimate final on filing, the estimate stands and the measurement is archived. The evidence fault under J1 stands, with whatever consequence the agreement attaches to it under A7.

A measurement filed after W+30 and by W+45, a first measurement or a correction, enters the record marked late and is archived; no pass for the Window uses it. A measurement this clause archives, here or under EF-07 above, has no effect on any interval's value, eligible denominator, fraction, materiality decision or conformance decision.

By W+30, a correction to a measurement already filed is filed under the W+30 row, and the first paragraph of this clause governs first measurements only. The correction leaves the measurement it corrects, that measurement's filing time and any evidence fault in the record, and Section H reads the counterparty's filing against the corrected figure as against any other.

J3 · After quantity finality. A measurement filed after W+45 is not part of the record and MUST NOT be used to compute the Period Figure.

J4 · Persistent failure to file. Where a Filing Party fails to file under H1 for more than the unresolved ceiling across a quarter's Windows, its Filing Plan is deemed not implemented for that quarter and I.2.1 applies.

J5 · Undeclared departure. A filing made under a Plan with an undeclared departure under G8 is not conforming, and every Period Figure resting on it is not conforming, for as long as the departure stood.

K · Custody

K1 · Retention. A Filing Party MUST retain the filed record, and the samples underlying it, for thirty-six months after money finality for the Window.

K2 · Access. Each Filing Party holds its own filings and receives the counterparty's filed set. Nothing in this standard requires a party to disclose telemetry beyond the samples it filed.

K3 · Onward disclosure. A Filing Party may give a third party access to the filed record under an access grant. An access grant names the party, the Windows, and its duration. A record disclosed under an access grant carries its signatures and inclusion proofs, and remains checkable in the hands of the grantee without either original party's cooperation.

K4 · Confidential terms. Where the agreement withholds a value this standard requires to be fixed — a threshold, a remedy schedule, a Grade Class dimension — the Filing Plan MUST record that the value is fixed, identify it by digest, and name who holds it. A withheld value that is committed before the window still binds. A withheld value that is not committed is not a standard, and a Period Figure resting on one is not conforming.

K4 is the whole of what confidentiality may cost. A party may keep its numbers private. It may not keep them movable.

K5 · Reliance by a grantee. Where a third party has taken an access grant under K3 and relied on a filing that carries a valid signature under D4 and an inclusion proof under E2, **the Filing Party that made that filing may not, as against that third party, rely on a fact contrary to it** — unless the Filing Party shows the grantee had notice of the filing's non-conformity at the time it relied.

L · Versions

A standard with no amendment procedure is a document. A standard whose amendments reach backwards is worse than none.

L1 · Identity. This standard is identified by number and version. A Profile, a Filing Plan and a conforming record each name the version they were armed under.

L2 · Lifecycle. Every metric registry entry under Annex 2.1 carries `class` (`canonical` or `experimental`), a `rolloutState`, and a `retirementPolicy`.

L3 · Effect. A new version takes effect at the opening of a Window and never inside one.

L4 · Non-retroactivity. A new version does not reach a Profile already armed under an earlier one. Retirement of an enumerated entry does not reach Windows already closed under it.

L5 · Permanent conformance. Conformance is to a named version, permanently. A later version never revokes conformance already established under an earlier one.

L6 · Defects found between versions. A defect in this standard is reported to PSYCHE and enters a public errata record. Each entry names the clause, the defect, and its disposition.

L6.1 A defect that changes what a clause requires is **substantive**, and it is corrected only by a new version under L3. **The text of a published version is never edited in place.** A10 exists so that a standard stops moving, and an errata process that rewrote a published clause would reopen the hole A10 closes.

L6.2 A defect that changes no requirement — a typographical error, a broken cross-reference, an arithmetic slip in an example — is **editorial**, and it is corrected by a dated erratum published against the version's digest and carried alongside it. The version's own digest is unchanged, and a record armed under that version stays armed under it.

L6.3 The errata record is published with the standard and is part of it. **A reader holding a version without its errata holds an incomplete document**, which is why L1 requires a Profile to name the version rather than to reproduce the text.

L7 · Amendment across armed Profiles. A Profile armed under one version continues under it, by L4. Where the parties to an armed Profile want a later change without waiting for their next arming, PSYCHE publishes the change as a digest-identified **Amendment Notice**, and the two Filing Parties adopt it by jointly signing an **Adherence** naming their existing Plan digests under G9 and the Notice's digest.

L7.1 An Adherence takes effect at the opening of a Window and never inside one, per L3, and MUST be committed before that Window under A10.

L7.2 An Adherence binds the two parties that signed it and reaches no other Profile. A Profile that has not adhered continues exactly as before. **Conformance already established is untouched**, because the Adherence changes what a future Window is armed under and leaves closed Windows to be read under the digest that governed them.

L8 · Continuity. L5 promises that a conformance question has one answer permanently, and I.2 contemplates a record read across the term of a facility. **Both promises outlive any single maintainer.** PSYCHE therefore deposits, with each version, the full text, the errata record, the schemas Annex 5 names, the registry entries Annex 2.1 indexes, and the material required to verify the digests, in a form that remains retrievable if PSYCHE does not. A version's deposit location is published with it.

The vocabulary and the permitted sets move. L5 is the claim that the question *was this record conforming under SQCD-1 v1.0* has one answer, and that answer does not change when v2 issues.

Annexes

Annex 1 · Filing Plan form

The fields of Section G, in order, as a form. A Plan filed on this form and signed under D1 satisfies G.

```
SQCD-1/P · FILING PLAN

G1  Filing party ................ [ legal name ]
    Role ........................ [ provider | customer ]
    Obligations covered ......... [ A1 references ]

G2  Grade Classes ............... [ per A8.1–A8.9, one block per class ]

G3  Instruments and Layers ..... [ per covered Layer: instrument,
                                   source, interval, clock, accuracy ]

G4  Collection and validation .. [ each step, each function ]

G5  Period function ............ [ Annex 2 number, or full statement ]

G6  Minimum observation ........ [ B8 for a customer; coverage for a
                                   provider ]

G7  Audit and test schedule .... [ cadence, party, standard ]

G8  Declared modifications ..... [ section, departure, reason ]
                                 [ "none" is a permitted entry ]

G9  Plan digest ................ [ algorithm, digest ]
    First Window governed ....... [ date ]

Signed ......................... [ D1 key identifier ]
```

Annex 2 · Enumerated sets

Every set below enumerates **forms**, not values. The Profile selects a form by number and supplies its parameters. Comparability across contracts comes from knowing which form was selected; it does not require that everyone select the same one.

Where a term allocates loss between the parties, this standard assigns no value, and for every such term but the required evidence class under A8.11 it enumerates what may be chosen.

2.1 · Period functions. A2 requires a function in this annex to be **named**, not described. A name resolves to an entry in the metric registry.

2.1.1 · Relationship to ISO/IEC 19086-2. A registry entry is a cloud service metric under **ISO/IEC 19086-2:2018, Clause 8**, the metric model of the 19086 SLA framework [19086-1], extended under that clause's own extension provision at 8.1.4. The table below maps the attributes Clause 8 defines onto the fields of `metric_registry_entry`.

Clause 8	`metric_registry_entry`
`Metric` identifier	`metricId`
description	`purpose`
source	`owner`
scale	`scale`
expression, with the `Expression` class	`algorithmReference`, which names the clause of this standard stating the function, and `outputUnit` for its unit
parameters and underlying metrics, with the `Parameter` class	`allowedInputs`
rules, with the `Rule` class	`numericRules`

`scale` is taken from 19086-2, and its binding there is preserved here: a cloud service level objective takes an interval or ratio scale, and a cloud service qualitative objective a nominal or ordinal one. **Every function in this annex is quantitative and takes an interval or ratio scale.** The Grade Class fields under A8 are specification values rather than period figures; A8.10 governs them, and no function in this annex applies to them.

2.1.2 · What this standard adds. Clause 8 specifies a metric well enough to describe a service and to compare two of them. Settlement asks three further questions, and a registry entry under this standard answers them.

Added to the 19086-2 entry	Why settlement requires it
`missingDataBehavior`	What the function does with an interval no sample reached. This value is exercised in Section C. A function that leaves it unstated has no answer for the case Section C exists for.
`determinismConstraints`	Whether two parties running the entry over one filed sample set are obliged to reach one figure, or, under 2.1.4, the same named quantities in its place. Comparability is enough to describe a service; settlement needs identity.
`version` · **`class`** · **`rolloutState`** · **`retirementPolicy`** · **`reviewRefs`** · **`profileScope`**	ISO/IEC 19086-2 carries no version field. A2 is committed under the A10 digest and read back under L1 through L5, and a metric able to change under a fixed name defeats every one of those clauses.

A metric a Window's figure was computed under must be recoverable years later, exactly as it then read.

2.1.3 · Where an entry lives. An entry is an object under the `metric_registry_entry` schema Annex 5 names. A Profile that names an entry MUST carry that entry, at the version named,

among the artifacts the A10 digest is computed over, so the record of a Window holds the entry as it read when the Window was armed and an evaluation resolves the name from that commitment and from nothing outside it. The index below is not such a carrier; the seven `canonical` entries it names are deposited with this version under L8, as `sqcd1-registry/`.

A Profile selects by identifier; an evaluation runs against the entry, not this table.

No.	Function
F-01	Time-weighted availability. Conforming interval-seconds divided by eligible interval-seconds in the Window, per Measured Subject, arithmetic mean across subjects.
F-02	Subject-weighted availability. As F-01, weighted by the accelerator count of each Measured Subject.
F-03	Worst-subject availability. The minimum of the per-subject F-01 values across the Window.
F-04	Aggregate availability. Conforming subject-seconds summed across all subjects, divided by eligible subject-seconds. Permitted only where the agreement names the aggregate as the Measured Subject under B3.
F-05	Delivered hours. The count of conforming Measured-Subject-hours in the Window, undivided.
F-06	Shortfall hours. Eligible Measured-Subject-hours less F-05.
F-07	Windowed availability with error budget. As F-01, with the first n non-conforming interval-seconds per subject per Window removed from both numerator and denominator, n declared in A5.

F-01 and F-04 over the same samples produce different months, and that is why A2 requires one of them to be named and A10 commits it before the Window opens.

An entry marked `experimental` MAY be selected by a Profile and MUST be recorded as experimental in the Filing Plan. Retirement of an entry does not reach Windows already closed under it; see Section L.

2.1.4 · A quantity a function cannot take. Where A4 excludes every interval of the Window, or a division the selected function performs would have a zero denominator, or a mean or a minimum it takes would run over an empty set the function requires, a per-subject domain the function needs included, no Period Figure is produced for the Window. A zero numerator, a zero reading and a zero minimum are computed as any other value. The evaluation returns `requires-review` and names every undefined quantity by obligation reference, Measured Subject identifier and quantity name, sorted in that order, each in code-point order, a quantity of the whole Window carrying an empty subject identifier. A quantity name is the registry entry's `allowedInputs` name for the input a zero denominator is formed from, `measured_subjects` where the set of subjects is empty, or `intervals_eligible_before_c5` where A4 excludes every interval. An evaluation MUST NOT put a value in place of the figure or remove a Measured Subject to obtain one. No Divergence is computed under H2, and no amount is computed under the remedy schedule. The Case proceeds under F2, or under I.1.1 where no Decision is filed, and what is owed follows the agreement's terms identified under A7; where the Case expires, the selected Annex 2.4 form applies as its row states.

For 4.3 the named quantities stand in place of the figure, and they recompute from the filed record. The record meets 4.6 when it states that no Period Figure was produced, names the undefined quantities as above, and records that the amount is not computed under this standard and follows the agreement's terms identified under A7. Where A4 excludes every interval, `intervals_eligible_before_c5` also stands in place of the Unresolved and Unilateral Fractions under 4.4; otherwise those fractions are stated as for any Window, and 4.4 and C5.3 read them as stated.

2.2 · Materiality threshold forms. The divergence at or below which two Filing Parties' Period Figures are treated as one. A6 requires a form and its parameters.

This set enumerates the forms the materiality threshold may take.

No.	Form
MT-01	Fraction of the Period Figure. Parameter: the fraction.
MT-02	Count of intervals on one Measured Subject. Parameter: the count, in intervals of the A3 sample interval. Permitted only where every Filing Plan in force samples at that interval; see G10.
MT-03	The greater of an MT-01 fraction and a quantity of Measured-Subject-hours. Parameters: the fraction, and the quantity. Permitted only with F-05 or F-06.
MT-04	The lesser of an MT-01 fraction and a quantity of Measured-Subject-hours. Parameters: the fraction, and the quantity. Permitted only with F-05 or F-06.
MT-05	Zero. Every non-zero Divergence is material and produces a Divergence Event.

MT-03 sets the wider band of the two and MT-04 the narrower.

2.2.1 · The comparison each form makes. Under MT-01 the fraction is taken of the Period Figure of the Filing Party the Profile names under A6, and the Divergence under H2 is compared with that product; under MT-05 it is compared with zero. Under MT-02 the comparison is made on each Measured Subject: the numbers of conforming intervals the two Filing Parties' Period Figures count on that subject are compared, and the Divergence is material where the difference on any one subject exceeds the parameter. Under MT-03 and MT-04 the Period Figure is a count of Measured-Subject-hours, the fraction is taken as under MT-01, and the Divergence under H2 is compared with the greater (MT-03) or the lesser (MT-04) of that product and the quantity. A Profile selecting MT-02 where a Filing Plan in force at arming samples at another interval, or MT-03 or MT-04 with a period function other than F-05 or F-06, cannot be armed, and G10 refuses a later modification that would set another interval under MT-02.

2.3 · Unresolved ceiling forms. The Unresolved Fraction above which the Period Figure is not conforming. A6 requires a form and its parameters.

This set enumerates the permitted forms of a threshold parameter of the armed parameter pack, `unresolved_ceiling`, which carries the selected form and its parameters.

No.	Form
UC-01	Fraction of the Window's intervals eligible before C5, in aggregate. Parameter: the fraction.
UC-02	Absolute count of intervals counted in the Unresolved Fraction. Parameter: the count.
UC-03	Fraction of intervals eligible before C5, evaluated **per Measured Subject**, the ceiling breached if any subject exceeds it. Parameter: the fraction.
UC-04	No ceiling. The Period Figure conforms at any Unresolved Fraction, **provided the fraction is stated**.

UC-01 and UC-03 diverge exactly where the aggregate hides a single blind subject. UC-04 is permitted because some parties will want it; it does not permit leaving the fraction unstated.

2.4 · Expiry pricing. What silence costs the silent party. I.1.1 requires a selection.

None of these produces `accepted`. The Case's terminal state under every entry below is `expired`. What moves is money under a term fixed before the Window, not consent. No entry in this set may be read as acceptance, and a Profile that describes one as acceptance is not conforming.

No.	Form
EX-01	No amount. Expiry closes the Case and nothing is owed on the claim either way.
EX-02	Invoice stands, remedy lapses. The underlying invoice remains payable on its own terms; the claim for remedy in this Case lapses unexercised.
EX-03	Remedy stands, uncontested. The amount computed under the remedy schedule is owed; the silent party spent its opportunity to contest. Where 2.1.4 leaves the Window without a Period Figure, EX-03 prices nothing, and what is owed follows the agreement's terms identified under A7.
EX-04	Stated cost. Expiry carries an amount payable by the silent party, fixed in advance and independent of the merits. Parameter: the amount.

EX-02 and EX-03 are symmetric and both are ordinary negotiated positions.

2.5 · Evidence-fault allocation. Who bears an interval on which a Filing Party did not file. J1 requires one form **per obligation**.

No.	Form
EF-01	Suspended. The interval carries no value and no amount is owed on it until a measurement covering it is filed under J2. An interval still suspended when the correction window closes at W+30 is treated from then on as under EF-05, and that treatment binds every later figure.
EF-02	Admitted. The interval counts as delivered.
EF-03	Denied. The interval counts as not delivered.
EF-04	Attributed. The interval is borne by the party C5.4 attributes it to; where it carries no observation, 2.5.2 applies.
EF-05	Excluded. The interval leaves both numerator and denominator and, where no Filing Party observed it, counts toward the Unresolved Fraction.
EF-06	Split. Borne in a declared ratio. Parameter: the ratio.
EF-07	Estimated. The interval carries a value computed from a formula named before the Window, over observations the Filing Party did produce. Parameters: the formula, and whether the estimate is replaceable by actual data or final on filing.

EF-07 answers a different question from EF-01 through EF-06. The first six allocate *who bears* an interval; EF-07 states *what number it carries*. Settlement-quality metering regimes this standard was drawn from do the second thing, at least one of them by a formula published in advance rather than by a judgment made afterward, and that one marks certain substitutes as final substitute data under its rules, with stated exceptions [AEMO]. J1 takes one form per obligation, so a Profile that wants an estimate for an obligation selects EF-07 for it in place of the other six. A Profile selecting EF-07 without stating which of the two it means has fixed nothing.

2.5.1 Conformance rule. A selection is not conforming where the party that holds the required evidence is the party that gains from its absence. EF-02 with a provider-held evidence class, and EF-01 or EF-03 with a customer-held one, are the standing instances. This is detectable when the Profile is armed and MUST be reported then.

2.5.1 refuses a selection on its merits only where a party is paid to stay blind.

2.5.2 An interval no party observed. Where C5.4 attributes an interval that carries no observation, the selected form assigns its value as follows, until a measurement the first paragraph

of J2 admits for the pass covers it. EF-01 suspends it, and from W+30 treats it as EF-05 does. EF-02 counts it as delivered and EF-03 as not delivered. EF-04 and EF-05 exclude it from numerator and denominator and count it in the Unresolved Fraction, the fault recorded against each party C5.4 names. EF-06 applies its ratio to whatever consequence the agreement attaches to the fault under A7, the interval itself excluded as under EF-05. EF-07 computes its value from the named formula.

2.6 · Substitution and adjustment forms. A9.2 forbids netting or substituting hours of different Grade Classes without an adjustment factor. These are the forms that factor may take.

No.	Form
SB-01	No substitution. Hours of different Grade Classes are separate obligations and never offset.
SB-02	Named-pair factor. A scalar for one ordered pair of Grade Classes, in one direction only. Parameters: the pair, the direction, the scalar.
SB-03	Field-derived factor. The ratio of one named A8 field between the two classes, that field stated. Permitted only where the obligation is measured on that field.
SB-04	Schedule. A table of factors over an enumerated set of class pairs, fixed before the Window.
SB-05	Upward only. Substitution permitted where every A8 field of the delivered class meets or exceeds the contracted class, at factor 1.

2.6.1 A factor MUST NOT be derived from price. A price ratio prices two goods; it does not establish that one discharges an obligation stated in the other.

2.6.2 SB-03 and SB-05 are checkable from the Grade Classes and, for SB-05, any ordering the agreement fixes under 2.6.3. SB-02 and SB-04 are negotiated numbers and MUST be fixed under the A10 digest before the Window.

2.6.3 · Comparison under SB-05. A field carrying a quantity, a count or a floor meets or exceeds where the delivered value is not below the contracted value; a version under A8.9 meets where it is not older. A field carrying a name or an enumerated value — `model`, `form_factor`, `memory_type`, `die_topology`, `link_generation`, `fabric_type`, `allocation_contiguity`, `region`, `availability_zone`, `power_region`, `placement_granularity`, `tenancy`, `preemptibility`, `isolation_attested` — meets only where the delivered value equals the contracted value, unless the agreement orders that field's values under the A10 digest before the Window. A field marked `not-material` in the contracted class is not compared.

SB-05 is the only form under which a provider may substitute with no factor to compute or negotiate, and a customer verifies it by reading the two Grade Classes and any ordering the agreement fixes under 2.6.3. Of the forms that permit substitution, it is also the only one that cannot be used to deliver less.

2.7 · Divergence resolution. How a material Divergence under Section H is carried to one quantity. H4 requires a selection.

None of these produces `accepted`. Every entry below decides how a **measurement** is carried to one quantity. `accepted` continues to come from F2 alone and `expired` from I.1.1, and a Profile that describes any entry here as acceptance is not conforming. Resolving a divergence settles the measured quantity; the counterparty's decision under F2 is a separate act by a party holding standing.

No.	Form
DR-01	No resolution. Both Period Figures stand in the record and the quantity is agreed by a route outside this standard.
DR-02	Tolerance mean. Where the two figures differ by no more than a stated bound, their arithmetic mean governs. Parameter: the bound.
DR-03	Third reading, closest two. A third reading is taken by a party named before the Window; the mean of the two closest of the three governs, and the outlier is discarded without adjudication. Where the three readings are equally spaced, the mean of all three governs. Parameters: who takes the third reading, and who bears its cost.
DR-04	Referee. A single referee named before the Window determines the quantity from the filed record. Parameters: the referee, and the cost rule.
DR-05	Panel. An odd-numbered panel drawn from a roster named before the Window decides by majority. Parameters: the roster, the panel size, and whether one appeal lies.
DR-06	Carried to decision. The divergence is carried into F2 unresolved, and the party with standing decides on a record that states both figures and the first interval at which they part.

2.7.1 A selection MUST be made. **DR-01 is a selection**, and a Profile that names it has chosen to leave the quantity to a conversation. Two parties may take that position deliberately, and this standard requires them to take it on the page rather than by omission.

2.7.2 DR-02 through DR-05 determine a quantity and MUST NOT be described as determining what is owed. The remedy schedule under A7 converts a quantity to money, and F2 settles, or, where no Decision is filed, the Annex 2.4 form under I.1.1.

2.7.3 Where the selected form requires a third party, that party MUST be named before the Window opens, together with the basis on which it is paid. A referee chosen after the divergence is known is chosen by whoever is ahead.

Physical commodity markets answer this question and have for a century: a mechanical two-of-three average, with the mean of all three where the three results are equally spaced [GAFTA124]; an assay the claimant pays for first, recovering that cost and the replacement cost where a deficiency is found, with arbitration still open [COMEX]; three graders drawn at random deciding by majority whether the goods meet the delivery standard, with one appeal to three others [ICE]; a third arbitrator appointed by the trade body to chair the tribunal [FOSFA]. Each of them ends in a stated measurement, a grade decision or an award. **What H4 contributes to that inheritance is the localisation** — the parties arrive at whichever of these forms they chose already knowing the interval at which their instruments parted, which none of those procedures supplies.

2.8 · Conformity decision near a limit. Whether a reading that sits close to a Grade Class floor or an A5 threshold counts as conforming [JCGM106]. Annex 7 requires a selection wherever a measured value is compared to a limit. Acceptance and rejection in this annex are conformity decisions on a reading [JCGM106]; a party's decision on a Claim is F2's.

A reading carries an expanded uncertainty under 7.0.2. Comparing it to a limit as though it were exact makes the outcome near that limit close to a coin flip, and which way the coin falls is worth money to one of the parties. **Which form applies allocates that risk, and this standard sets no default.**

No.	Form
CD-01	Simple acceptance. The reading is compared to the limit directly and the risk arising from measurement uncertainty falls where it falls.
CD-02	Guarded acceptance. The reading must clear the limit by at least a stated guard band before the field counts as conforming. Parameter: the band, stated as a multiple of the expanded uncertainty U.
CD-03	Guarded rejection. The reading must miss the limit by at least a stated guard band before the field counts as non-conforming. Parameter: the band.
CD-04	Stated maximum risk. The band is derived so that the probability of a wrong conformity decision does not exceed a stated value. Parameter: the maximum risk.

2.8.1 A form MUST be selected for every Grade Class field measured under Annex 7 against a floor, and for the A5 threshold where the Period Figure is built from measured values rather than from counted intervals.

2.8.2 CD-02 protects the party that loses by a wrong acceptance; CD-03 protects the party that loses by a wrong rejection. **They are the same instrument pointed in opposite directions**, and a Profile selecting one has decided which error it would rather make.

2.8.3 Where a selection names a guard band as a multiple of U, the coverage factor k used to compute U MUST be the one the procedure states under 7.0.2. A band expressed against an unstated k is a band of unknown width.

Annex 3 · Worked example

One Window, one obligation, two Filing Parties. Every number below is derived from the terms in 3.1 and the sample counts in 3.2 and can be recomputed from them with no other input.

What this example does not exercise. It demonstrates counted intervals and a period function. **No reading under Annex 7 appears in it.** The Period Figures below are computed exactly from counts of conforming and eligible intervals, and the fractions are counts of intervals divided by those eligible before C5 — exact ratios of integers — and they carry no expanded uncertainty because no instrument produced them. A Window whose Period Figure rests on measured values instead of counted intervals runs 7.0.2 for every reading and 7.0.2.1 for every comparison against a floor, and its figure would be reported with the uncertainty those clauses require. **The six decimal places below are the arithmetic of a division.** A standard that publishes rules and withholds arithmetic leaves every implementer to guess at the same question.

3.1 · The standard in force.

Ref	Term	Value
A1	Obligation	Availability of contracted capacity, clause 4.2
A2	Period function	**F-01**, time-weighted availability, per subject, arithmetic mean across subjects
A3	Window	2026-06-01T00:00:00Z → 2026-07-01T00:00:00Z. 720 hours. Sample interval 300 s. Samples straddling either edge are assigned to the Window containing their start.
A4	Exclusions	Announced maintenance, notified ≥ 72 h in advance. Excluded intervals are removed from **both** numerator and denominator.
A5	Tolerance / threshold	No error budget. Obligation missed below 99.90 %.
A6	Materiality	**MT-02**, count of intervals on one Measured Subject. Parameter: **12 intervals** (one hour). Both Filing Plans sample at the A3 interval of 300 s.
A6	Materiality reference	the provider's Period Figure, named under the A10 digest
A6	Unresolved ceiling	**UC-01**, fraction of intervals eligible before C5, in aggregate. Parameter: **1 %**.
A7	Remedy schedule	≥ 99.90 % → none · 99.85 % to below 99.90 % → 5 % · 99.50 % to below 99.85 % → 10 % · < 99.50 % → 25 %, of the Window fee
A8	Grade Class	`GC-H200-SXM-141`, 8 accelerators per Measured Subject
A10	Standard digest	committed 2026-05-28, before the Window opened
A8.11	V4 declarations	`tenancy = dedicated`, `allocation_contiguity = contiguous`, both signed by the provider 2026-05-28, under the A10 digest; `preemptibility` and `placement_granularity` marked `not-material`
A9	Substitution	**SB-01**, no substitution
B3	Measured Subject	Node. Eight subjects, N-01 … N-08
B7	Covered Layers	L2, L3, L4, L6, L7. L1 and L5 out of scope. **L8 never counts.**
B8	Minimum customer observation	Reachability of each subject at 300 s, plus the customer's own record of work submitted, started and completed
Annex 2.5	Evidence-fault allocation	**EF-04**, borne by the party C5.4 attributes it to
H4	Divergence resolution	**DR-06**, carried to decision
I	Timetable	W+2 measurements · W+5 divergence · W+7 Claim · W+12 Decision · W+30 correction closes · W+45 quantity final · W+90 money final
I.1.1	Expiry pricing	**EX-02**, invoice stands, remedy lapses
—	Price	$2.90 per accelerator-hour

Window fee: 8 subjects × 8 accelerators × 720 h = **46 080 accelerator-hours** × $2.90 = **$133 632.00**

3.2 · What the instruments recorded. Intervals per subject: 720 h × 3600 ÷ 300 = **8 640**. Total subject-intervals: 8 × 8 640 = **69 120**.

#	Event	Extent
E1	Staged firmware maintenance, notified 2026-05-25. **A4 exclusion.**	24 h = 288 intervals, on N-01 ... N-04 → **1 152** subject-intervals
E2	Both collectors lost their upstream, and each party filed a BoundaryRecord for the interval with cause `instrument-down`. **No Filing Party observed; both declared.**	45 min = 9 intervals, all subjects → **72** subject-intervals
E3	Customer probe down. **Provider filed, and its record is conforming throughout; customer filed nothing and declared no gap.**	4 h = 48 intervals, all subjects → **384** subject-intervals
E4	N-03: control plane could not place work. **L3 healthy throughout; L7 non-conforming.** Both parties filed, and agree.	6 h = 72 intervals on N-03
E5	N-05: customer probe records unreachable at L6; provider's collector records conforming. **The parties disagree.**	3 h = 36 intervals on N-05, from 2026-06-21T02:00:00Z

The five events cover disjoint intervals: no subject-interval falls in two of them. Every other subject-interval was observed by both Filing Parties and is conforming at every covered Layer.

E4 is the case B7.2 exists for. Every accelerator on N-03 was healthy and at its declared clocks for all 72 intervals. The obligation covers L7. An interval is delivered only where every covered Layer was simultaneously conforming, so the 72 intervals are non-conforming and a reading at L3 is not an answer about them.

3.3 · Denominators. Excluded (E1) is removed from both sides. Unresolved (E2), declared by both parties, is removed from both sides under C5.2; had either party left its gap undeclared, C5.4 would have attributed the interval to it.

	Intervals
Total subject-intervals	69 120
less A4 exclusions (E1)	− 1 152
Eligible before C5	**67 968**
less Unresolved (E2)	− 72
Eligible	**67 896**

Per subject: N-01 ... N-04 → 8 640 − 288 − 9 = **8 343**. N-05 ... N-08 → 8 640 − 9 = **8 631**. Check: 4 × 8 343 + 4 × 8 631 = 33 372 + 34 524 = 67 896. ✓ Every subject's eligible count is above zero, so Annex 2.1.4 does not apply.

Unresolved Fraction = 72 ÷ 67 968 = **0.105932 %**. Ceiling 1 % (UC-01). Conforming.
Unilateral Fraction = 384 ÷ 67 968 = **0.564972 %**. No ceiling applies; it is disclosed.

E3 is unilateral: only the provider observed. The customer's Filing Plan required that observation under B8, and the customer neither produced it nor declared a gap, so C5.4 applies and J1 routes to the Profile's Annex 2.5 selection. Under **EF-04** the interval is borne by the customer, and the provider's record — conforming — determines all 384.

3.4 · The two Period Figures. Per-subject availability, conforming ÷ eligible, shown to eight decimals; the Period Figures below use the exact ratios:

Subject	Provider	Customer
N-01, N-02, N-04	1.00000000	1.00000000
N-03	$1 - 72/8\,343 =$ **0.99137001**	**0.99137001**
N-05	1.00000000	$1 - 36/8\,631 =$ **0.99582899**
N-06, N-07, N-08	1.00000000	1.00000000

F-01, provider $= (8\,271/8\,343 + 7) \div 8 =$ **99.892125 %**
F-01, customer $= (8\,271/8\,343 + 8\,595/8\,631 + 6) \div 8 =$ **99.839987 %**

Divergence = 0.052138 percentage points. All of it arises from 36 intervals on N-05.

3.5 · Divergence. Materiality is MT-02 at 12 intervals on one Measured Subject, and both Filing Plans sample at the 300 s of A3, so the comparison under 2.2.1 runs subject by subject. On N-05 the provider's figure counts 8 631 conforming intervals and the customer's 8 595, a difference of 36; on every other subject the two counts are equal. **36 > 12 → material.** A Divergence Event is filed under H4, by arithmetic, without either party requesting it.

H4 field	Value
Both Period Figures	99.892125 % · 99.839987 %
Interval decomposition	filed, per subject, per interval
First interval at which the series part	**2026-06-21T02:00:00Z**
Layer and subject	**L6**, N-05
Unresolved Fraction, each party	0.105932 % · 0.105932 %

The evidence set is now known to be inconsistent. Under DR-06 the evaluation does not choose between the two readings; it records the conflict and carries it forward, and returns `requires-review` rather than a support assertion. Choosing a winner here would be the evaluation exercising an authority it does not hold.

Locating the disagreement to 2026-06-21T02:00:00Z is most of the work. The parties are no longer arguing about a month. They are arguing about three hours on one node, with both instruments' readings of those three hours in front of them.

3.6 · The money.

	Period Figure	Tier	Credit
Provider's filing	99.892125 %	99.85 % to below 99.90 %	5 % → **$6 681.60**
Customer's filing	99.839987 %	99.50 % to below 99.85 %	10 % → **$13 363.20**

The disputed amount is $6 681.60, and it is 36 five-minute intervals on one node out of eight, in a 720-hour month.

Both quotes are computed and both are carried. **Neither is owed yet.** The remedy quote names an amount; the amount owed is fixed when the party with standing signs a decision under F2, or, where none is filed by W+12, by the expiry form selected under I.1.1, here EX-02: the invoice stands and the remedy lapses. Under DR-06 the quantity is decided with F2, so neither the measured quantity nor the Counterparty Decision is settled yet.

3.7 · The same samples under a different declared function. A2 required the function to be named, and A10 committed it before the Window opened. The table below shows why.

Function	Provider's figure	Tier	Credit
F-01 time-weighted, mean across subjects	99.892125 %	99.85 % to below 99.90 %	5 % → **$6 681.60**
F-04 aggregate, subject-seconds pooled, with B3 naming the aggregate	99.893955 %	99.85 % to below 99.90 %	5 % → $6 681.60
F-03 worst subject	**99.137001 %**	**< 99.50 %**	**25 % → $33 408.00**

Identical samples and identical instruments produce **five times the money**.

F-01 and F-04 differ here in the third decimal place, because the eligible denominators differ only by E1's staged maintenance. F-03 differs by three quarters of a percentage point, because it reports the worst subject rather than the fleet, and one node's control plane could not place work for six hours.

All three are correct. **The one that is wrong is the contract that did not say which.**

3.8 · Completeness. Give this record to someone who was absent and holds no account on either side.

	Comes out
The period figure	99.892125 % and 99.839987 %, each recomputable from 3.1, 3.2 and 3.3
The shortfall	72 non-conforming intervals agreed, 36 disputed, located to the first interval at which the series part
The amount owed	$6 681.60 or $13 363.20, quoted under A7, **fixed when the party with standing signs a decision under F2**, or, where none is filed by W+12, by EX-02, under which the remedy lapses
What was not observed	Unresolved 0.105932 %, Unilateral 0.564972 %, both disclosed
What was not verifiable	`tenancy` and `allocation_contiguity` are V4. Neither was measured. Both were declared by the provider on 2026-05-28 under the A10 digest, and the declarations are in the record.

Each traces to a named field.

Annex 4 · Grade Class registry

An index of manufacturer-published specifications, each entry naming its source and the date it was read. Values are transcribed as printed. A field the manufacturer states only as a system total is marked **ABSENT** rather than divided, and a field for which the manufacturer prints two irreconcilable figures is marked **ABSENT** with both recorded.

Listing is citation rather than certification, and no manufacturer has undertaken anything to PSYCHE by appearing here. Under the V1 procedures in Annex 7, a customer establishes that each V1 field of the part in front of it matches.

This annex lists Grade Classes by name, with their A8 values, and PSYCHE maintains it.

Only verified values are printed. A field the manufacturer's page does not state is marked ABSENT and left empty. Arithmetic performed on printed values is not a printed value and is not entered here.

Device fields (A8.1, A8.2 where the device fixes it) come from the manufacturer. Deployment fields — `coherent_domain_size` where the rack fixes it, `fabric_*`, `allocation_`

`contiguity`, `tenancy`, storage and network floors — are properties of a deployment, not of a device, and are supplied per Profile.

GC-H200-SXM-141. Source: nvidia.com/en-us/data-center/h200/ · retrieved 2026-09-02 · quotation

Field	Value
`model`	NVIDIA H200
`form_factor`	SXM
`memory_capacity`	141 GB
`memory_type`	HBM3e
`memory_bandwidth`	4.8 TB/s
`link_bandwidth_per_gpu`	900 GB/s
PCIe	Gen5, 128 GB/s
TDP	up to 700 W (configurable)

GC-H200-NVL-141. Same source.

Field	Value
`model`	NVIDIA H200 NVL
`form_factor`	PCIe dual-slot, air-cooled
`memory_capacity`	141 GB
`memory_type`	HBM3e
`memory_bandwidth`	4.8 TB/s
`link_bandwidth_per_gpu`	900 GB/s per GPU
PCIe	Gen5, 128 GB/s
TDP	up to 600 W (configurable)

GC-GB300-NVL72. Source: nvidia.com/en-us/data-center/gb300-nvl72 · retrieved 2026-09-02 · quotation

Field	Value
configuration	72 NVIDIA Blackwell Ultra GPUs, 36 NVIDIA Grace CPUs
GPU memory, total	20 TB HBM3E
fast memory, total	37 TB
NVLink bandwidth, total	130 TB/s
networking	800 Gb/s (NVIDIA ConnectX-8 SuperNIC)
`memory_capacity` per accelerator	**ABSENT** — the page states the total only
`link_bandwidth_per_gpu` · rack power	**ABSENT**

GC-H100-SXM-80. Source: nvidia.com/en-us/data-center/h100/ · retrieved 2026-09-02, memory type re-read 2026-11-09 · quotation

Field	Value
`model`	NVIDIA H100
`form_factor`	SXM
`memory_capacity`	80 GB
`memory_type`	**ABSENT** — the page states capacity and bandwidth for this part and names no memory type for it
`memory_bandwidth`	3.35 TB/s
`link_bandwidth_per_gpu`	900 GB/s
PCIe	Gen5, 128 GB/s
TDP	up to 700 W (configurable)

GC-H100-NVL-94. Same source.

Field	Value
`model`	NVIDIA H100 NVL
`form_factor`	PCIe dual-slot, air-cooled
`memory_capacity`	94 GB
`memory_type`	HBM3
`memory_bandwidth`	3.9 TB/s
`link_bandwidth_per_gpu`	600 GB/s
PCIe	Gen5, 128 GB/s
TDP	350–400 W (configurable)

The manufacturer's current page prints no 80 GB PCIe H100. **`GC-H100-PCIe-80` is therefore not registered**, and a contract naming one is naming a part this registry cannot resolve.

GC-B200-SXM. Sources: nvidia.com/en-us/data-center/hgx/ and nvidia.com/en-us/data-center/dgx-b200/ · retrieved 2026-09-02 · quotation

Field	Value
`model`	NVIDIA Blackwell, B200
`form_factor`	SXM
`link_bandwidth_per_gpu`	1.8 TB/s, GPU-to-GPU
`memory_type`	HBM3e (DGX B200)
`memory_capacity`	**ABSENT** — eight-GPU totals only: 1.4 TB (HGX B200), 1,440 GB (DGX B200)
`memory_bandwidth`	**ABSENT** — 64 TB/s (DGX B200) is an eight-GPU aggregate
`coherent_domain_size`	**ABSENT** — the board is printed as 8× Blackwell SXM with 2× NVSwitch and 14.4 TB/s aggregate NVLink
networking	up to 400 Gb/s per port (DGX B200) · 0.8 TB/s system networking bandwidth (HGX B200)

Read the B200 entry against the two H200 entries. For the older part the manufacturer prints per-accelerator memory capacity and per-accelerator bandwidth. For the current part it prints eight-GPU totals, and the two fields a delivery obligation is actually written in have no printed per-accelerator value at all.

Dividing the total by eight would supply them, and this registry declines to, because a quotient is not a quotation and a Grade Class that mixes the two stops being checkable against anything. For a part whose manufacturer states them only as a system total, `memory_capacity`

and `memory_bandwidth` therefore reach the record as a Profile declaration under A8, and `memory_bandwidth` also as a reading under **MP-01**. The registry cannot supply them, and the standard's answer to a field the datasheet leaves out is the same as its answer everywhere else — measure it, or have the party that knows sign it.

GC-MI300X · GC-MI325X · GC-MI350X · GC-MI355X. Source: the single-accelerator product page and the platform page for each part at amd.com, each cross-checked against its product brief or datasheet PDF · retrieved 2026-09-02 · quotation

	MI300X	MI325X	MI350X	MI355X
`form_factor`	OAM	OAM	OAM Module	OAM
`memory_capacity`	192 GB	256 GB	288 GB	288 GB
`memory_type`	HBM3	HBM3E	HBM3E	HBM3E
`memory_bandwidth`	5.3 TB/s	6 TB/s	8 TB/s	8 TB/s
Infinity Fabric links	8, at 128 GB/s per link	7, at 128 GB/s per link	7, per-link rate **ABSENT**	7, per-link rate **ABSENT**
`link_bandwidth_per_gpu`	896 GB/s	896 GB/s	**ABSENT**	**ABSENT**
host	PCIe 5.0 x16, 128 GB/s	PCIe Gen 5, 128 GB/s	PCIe 5.0 x16, 128 GB/s	PCIe 5.0 x16
TDP	up to 750 W	**ABSENT**	1000 W typical board power	**ABSENT**
`coherent_domain_size`	**ABSENT**	**ABSENT**	**ABSENT**	**ABSENT**
`fabric_bandwidth_per_gpu`	**ABSENT**	**ABSENT**	**ABSENT**	**ABSENT**

`link_bandwidth_per_gpu` is unresolved for two of these four. Each platform page prints a figure labelled *Total Aggregate Bi-directional I/O Bandwidth* (*Peer-to-Peer*) — 896 GB/s for MI300X and MI325X, 1,194.8 GB/s for MI350X and MI355X — and the label says neither per accelerator nor per platform. For the first pair the manufacturer's own datasheet resolves it: it states seven links reaching the other seven accelerators at 128 GB/s between each pair, and $7 \times 128 = 896$ exactly, so the printed figure is a per-accelerator aggregate. **For the second pair nothing in the cited documents resolves it.** Their product brief prints 153.6 GB/s per link in its table and 160 GB/s per link in its prose, and neither figure times seven reaches 1,194.8. The field stays **ABSENT** for MI350X and MI355X.

GC-TPU-V5E · GC-TPU-V5P · GC-TPU-V6E · GC-TPU-V7. Source: docs.cloud.google.com/tpu/docs · retrieved 2026-09-02 · quotation. **The manufacturer labels per-chip and per-pod figures itself**, and the per-chip rows below are its own.

	v5e	v5p	v6e	v7
HBM per chip	16 GB	95 GiB	32 GB	192 GiB
HBM bandwidth per chip	800 GiBps	2,765 GBps	1,638 GBps	7,380 GBps
Interconnect per chip, bidirectional	400 GBps	1,200 GBps	800 GBps	1,200 GBps
chips per pod, maximum	256	8,960	256	9,216
topology	2D torus	3D torus	2D torus	**ABSENT**
power	**ABSENT**	**ABSENT**	**ABSENT**	**ABSENT**
`form_factor` · `coherent_domain_size`	**ABSENT**	**ABSENT**	**ABSENT**	**ABSENT**

The unit this manufacturer's documentation specifies is a chip inside a pod, and "chips per pod" is a scheduling maximum rather than a set of accelerators mutually reachable at link bandwidth. **`coherent_domain_size` has no printed value for any generation**, and a Profile contracting for these parts fixes it by declaration under A8 rather than by reference to this registry.

GC-GAUDI3. Source: the manufacturer's HL-325L OAM product brief · retrieved 2026-09-02 · quotation

Field	Value
`model`	Intel Gaudi 3, HL-325L
`form_factor`	OCP Accelerator Module (OAM) v2.0
`memory_capacity`	128 GB
`memory_type`	HBM2e
`memory_bandwidth`	"total throughput of 3.7 TB/s"
networking	24 × 200 GbE RoCE v2 RDMA ports; 9.6 Tb/s bidirectional
host	PCIe Gen 5.0 x16
TDP	up to 900 W
`link_bandwidth_per_gpu` · `fabric_bandwidth_per_gpu`	**ABSENT** — the brief states the 24-port fabric as one pooled figure and does not separate intra-node from inter-node
`coherent_domain_size`	**ABSENT**

GC-TRN2 · GC-TRN3 · GC-INF2. Source: the manufacturer's accelerator architecture documentation, its Trainium and Inferentia product pages, and EC2 instance pages · retrieved 2026-09-02 · quotation

	Trainium2	Trainium3	Inferentia2
`memory_capacity`	96 GiB	**ABSENT**	32 GB
`memory_type`	HBM	HBM3e	HBM
`memory_bandwidth`	2.9 TB/s	4.9 TB/s	820 GB/s
`link_bandwidth_per_gpu`	**ABSENT**	**ABSENT**	**ABSENT**
`coherent_domain_size`	**ABSENT**	**ABSENT**	**ABSENT**
`form_factor` · host interface · power	**ABSENT**	**ABSENT**	**ABSENT**

`link_bandwidth_per_gpu` is ABSENT for all three, because in each case the manufacturer prints two different figures for it. The chip documentation gives 1.28 TB/s per chip for Trainium2 and the instance table gives 1,024 GB/s per chip for what reads as the same quantity; Trainium3 is 2.56 TB/s per device in one document and 2 TB/s per chip in another;

Inferentia2 is 384 GiB/s per chip in the architecture table and 192 GB/s on the product page. **None of the manufacturer documents cited reconciles any of the three pairs.** Both figures are recorded here and neither is entered. Trainium3's `memory_capacity` is ABSENT on the same ground: 144 GiB per chip in the architecture documentation, 144 GB on the product and instance pages.

This manufacturer prints per-chip specifications and sells instances. **The documents cited print no form factor, no host interface and no power figure for any of these parts**, which follows from the manufacturer selling them as instances rather than as cards a customer installs. Its pooling language is scoped to the sixteen chips of one instance and is not extended to the Trn2 UltraServer, which connects 64 chips across four instances; whether that server is one pooled domain or four linked ones is left unstated. Either is a property of a deployment, supplied per Profile under A8.

Annex 5 · Record schemas

The canonical form of every object in the left column of the table below is a published schema. **The schemas are those the reference implementation runs on; this standard does not define a second set.**

Canonicalization and signature follow the implementation: canonical JSON under the digest suite `psyche.digestsuite/1` — keys sorted, compact separators, ASCII-escaped — digested with SHA-256, and Ed25519 signatures.

This standard	Schema
F1 Delivery Claim	`claim`: the window is `claim_window`; the standard is `profileId` with `registrySnapshotDigest`; the subject is `subject`; the pass under I.1.4.1 is `pass_identifier`, absent for the single pass in I.1; the value at risk is `value_at_risk`, an amount of money in `minor_units` of `currency`. The quantity is the one carried forward for that pass under H3, or under H4 and Annex 2.7, recomputed under 4.3; where the Annex 2.7 path has not yet decided it, the Claim names the Filing Parties' Period Figures for that pass as candidates, each recomputed under 4.3 with its basis. The evidence set is every `evidence_receipt` whose `claim_ref` names the Claim. A credit Claim under I.1.5 names the Claim it adjusts in its own `claim_id`, which is that Claim's `claim_id` followed by `/credit/` and the credit Claim's `pass_identifier`; a resubmission after a withdrawn credit Claim appends `/` and a sequence number, 2 for the first resubmission and one more for each after it, and the withdrawn credit Claim it follows carries the same identifier with the number one lower, or with no number where the resubmission's number is 2. Read from the right, the identifier parses one way: an optional sequence number of digits, the pass identifier as W+ and digits, the token `credit`, and the adjusted Claim's `claim_id` as the remainder. Its `value_at_risk` is the amount I.1.5 gives it
F2 Counterparty Decision	`claim_review`
Terminal state	`terminal_event` (`accepted` · `rejected` · `disputed` · `expired` · `withdrawn`)
Evaluation output	`sigma_output` (`claim-supported` · `claim-not-supported` · `evidence-fault` · `requires-review`)
C1–C5 Missing Intervals	`boundary_record`
B measurements, filed	`evidence_receipt` · `evidence_obligation`
A10 standard digest	`digest_suite` over the profile and its version, `profile_recipe` · `parameter_pack` · expected model · spot-check policy
A2 period function	`metric_registry_entry` — a 19086-2 Clause 8 metric plus the Annex 2.1.2 fields
A6 thresholds	`parameter_pack`
E2 inclusion proof	`settlement_head`, the reference implementation's signed head over a Case's settlement log. A filing's proof against it is `leaf_index`, `tree_size` and `path`: the audit path from the filing's leaf, the signed event canonically serialised, to `root_hash` at that `tree_size`, computed with the Merkle algorithm of [RFC6962] §2.1 and verified as [RFC9162] §2.1.3.2 verifies, over the reference implementation's own canonical event and head formats · `oracle_log_head`, the head of its spot-check issuance log, which binds by `oracle_log_head_digest`. A spot check proves membership by `leaf`, `index`, `leaf_count`, `audit_path` and `head`, where `head` is the root that digest names and each parent is the digest of `{"node": [left, right]}` under the suite above, an odd last node paired with itself. A settlement-log event binds `oracle_log_head_digest`: the Capsule submission, or an `expired` or `withdrawn` terminal event. The issuance log head's time bound is the time of the published and verified `settlement_head` whose tree includes that event; the event's own `occurred_at` is a time as claimed under E1
F3 authority reference	`grant`, the grant F2 was signed under, named by `claim_review.grant_id`; `authorization_snapshot` is the payer's authorization the release of money is checked against, not the review authority
Capsule and replay	`capsule`, with the verification material it is replayed from
Registry	`registry_snapshot` · common types

A record that validates against these schemas is not thereby conforming. Conformance is Section 4. **A record that does not validate against them is not conforming under any reading.**

Annex 6 · The evidence behind V1 and V2

A8.10 places some fields at V1 on the strength of a measurement the customer takes itself, and others at V2 on the strength of a signature a manufacturer stands behind. This annex states what each rests on, what each declines to carry, and what a Profile relying on either must record.

6.0 · The roles, and why they are borrowed rather than invented. Attestation has a standard vocabulary and this standard uses it [RATS]. An **Attester** produces Evidence about itself. A **Verifier** appraises that Evidence against Reference Values under an appraisal policy and produces an Attestation Result. A **Relying Party** consumes the Result and acts on it. An **Endorser** — in practice the manufacturer — vouches for the Attester's capability, and a **Reference Value Provider** publishes the values the Verifier compares against.

Mapped onto this standard: for a V2 field the Attester is the device, the Endorser and the Reference Value Provider are the manufacturer, the Verifier is whatever appraises the report, and the evaluation is the Relying Party. **For a V1 field the customer is Attester and Verifier at once** — which is why Annex 7 must state the appraisal policy explicitly. 7.0.1's declared conditions and the bounds in 7.1 are that policy, written out.

6.0.1 · Where the borrowed model runs out. That architecture assumes one Attester per fact. **This standard has two Filing Parties producing evidence about the same fact and disagreeing.** Section H exists for that situation, and the attestation architecture does not model it. The vocabulary is borrowed for the roles; the divergence machinery is this standard's own.

6.0.2 · Two topologies, and why the choice is filed. A Verifier may return its Result to the Attester, which presents it onward (the passport model), or it may sit behind the Relying Party and be consulted when the record is appraised (the background-check model). **The first survives the loss of the verification service. The second does not.** K1 requires the filed record to be retained for thirty-six months after money finality, and a Profile that depends on a vendor-hosted verifier at read time has made its record depend on that service still answering years later. **The Filing Plan MUST state which topology produced a V2 field**, and a record produced under the second SHOULD carry the Attestation Result itself rather than a reference to it.

6.1 · The V1 primitive. The source. [ATC], version 1, submitted 2026-06-22. A CUDA probe measures a streaming-multiprocessor-by-memory-region latency matrix using physical SM labels and dependent global loads. A reducer commits sufficient statistics, configuration, code hashes, network evidence and a compressed raw archive into a certificate **a verifier checks without a GPU**. Retrieved 2026-09-02.

What it reports, and the device each figure was taken on.

Claim	Reported	Device
Fingerprint stability	median temporal jitter 0.09 cycles over a six-hour full-load run	**RTX 5090**
Die separation	100.0 %, shape-normalised leave-one-out nearest-neighbour classifier	Blackwell dies
Topology-class recovery	unified memory domain · two-way L2 split · two-die package at a 30-cycle cross-die penalty	V100 · H200 · B200
Location	within 44 km of the claimed datacentre, on 168 of 169 responding probes after the consensus fit	B200
Location	within 270 km of the declared site, on 167 landmarks	V100

The stability figure was taken on a consumer part; the topology and location figures were taken on datacentre parts. **A standard that quotes 0.09 cycles against an H200 quotes a number nobody has measured on an H200.**

The decoy result, at both of its stages. Of eleven named decoy datacentre locations, round-trip time alone rejects **nine**. All eleven are rejected after the consensus fit, which discards a

probe reporting a physically impossible sub-millisecond round trip from a distant registered location.

A location test stopped at the first stage admits two false sites in eleven. MP-07 requires the second stage, and 7.1.6 states it.

A primitive that can locate is a convenience; a primitive that can **refuse a false claim** is evidence.

What the source declines to claim. The paper states that the certificate proves consistency of the measured evidence, and that it does not prove firmware integrity, prevent refusal to run, or replace vendor attestation where a trusted enclave is available. A host with physical control can cut power, refuse to run the probe, or avoid reachable services.

Refusal is a case this standard already holds. A probe the provider will not let run produces **no reading** under 7.0.3, enters the record as a BoundaryRecord under Section C, and is valued under C5.5 — the treatment any other absent evidence receives. **V1 means that misreporting requires defeating a physical measurement.** Preventing the measurement is a different act, and it resolves as an account of what was not observed rather than as a reading.

Independent precedent. The technique has a peer-reviewed lineage this standard did not draw on and should name. Published work on remote GPU fingerprinting separates physically identical units of the same model by stalling execution units and measuring the resulting timing skew — unprivileged, in a browser, across more than two thousand devices and sixteen hundred distinct GPUs [DA]. **It corroborates the physics rather than the datacentre result**; the parts, the conditions and the purpose all differ. It establishes that individual-die timing variance is a real and exploited signal rather than a claim resting on one unreplicated paper.

Status. The construction has **not** been reproduced by this standard's maintainer. Until it is, the three fields stand at V1 **on an existence result** rather than on a deployed evidence class, and a Profile relying on them MUST record that under G8.

Boundary. The primitive establishes identity, class and coarse location. It reaches no further: exclusivity, contiguity across accelerators, bandwidth floors and firmware integrity stay where A8.10 puts them.

What remains owed. Two things remain owed: independent reproduction on datacentre parts, the stability figure included, since that one was measured on a consumer card; and a false-accept rate for the landmark test. The source reports none, and reports one instance in which a second site remained acceptable alongside the correct one until colocation and autonomous-system evidence resolved it. Until a rate exists, MP-07 yields a coarse geographic position and a decoy rejection, and it carries no stated error.

Annex 7 · Measurement procedures

A8.10 places a field at **V1** when the customer can establish its value by physical measurement. This annex states what is required, because a value that two parties measure differently is not a measurement.

7.0 · What makes a reading admissible.

7.0.1 · Declared conditions. A reading carries the conditions under which it was taken: the load state of the subject, whether the measurement window was exclusive, the driver and runtime versions in force, and the wall-clock start. **A reading whose conditions are not declared is not admissible**, whatever its value.

7.0.2 · Uncertainty. A reading is the median of n runs, and it is reported with the uncertainty of that median rather than as a bare number.

Let s be the median absolute deviation of the runs scaled by 1.4826, and let $u = 1.253 \cdot s \,/\, \sqrt{n}$ be the standard uncertainty of the reported median. The **expanded uncertainty** is $U = k \cdot u$ [GUM], where k is the coverage factor the procedure names. **Where a procedure names its bound as U/median, a reading is admissible only where U divided by the median falls inside that bound, at the coverage factor the procedure states.**

The scaled median absolute deviation is a robust estimator of the standard deviation, and the factor 1.253 relates the uncertainty of a median to that of a mean for runs consistent with a single normal distribution. A procedure MAY name a different estimator; it MUST name the one it uses.

A single number with no uncertainty is not a measurement. It is an assertion with a unit attached. Here this standard imposes a numeric requirement without the Profile's choice: an instrument's uncertainty is a property of the instrument, and allocates nothing between the parties. **What that uncertainty then decides is a different question, and 7.0.2.1 hands it back to the Profile.**

7.0.2.1 · Conformity near a limit. Where a reading is compared to a Grade Class floor or to the A5 threshold, the comparison uses the form the Profile selects from **Annex 2.8**. **SQCD-1 sets no default.**

7.0.2.2 · Two parties, one field. 7.0.2 bounds the agreement of a reading with itself. It says nothing about the agreement of two Filing Parties' independent readings of the same field. Section H is about that quantity.

Where both parties read the same V1 field, the Profile MUST state the **reproducibility limit** — the largest difference between two independent readings, taken by different parties on different instruments, to be treated as measurement rather than as divergence. **A materiality threshold under A6 applied to measured fields without a reproducibility limit turns instrument scatter into a Divergence Event**, and the parties then litigate their own noise.

7.0.2.3 · Status of the bounds in 7.1. The bounds this standard names in 7.1 for MP-01 through MP-05 are stated as relative expanded uncertainty at $k = 2$. **They rest on no interlaboratory study.** A reproducibility trial across independent operators and instruments would make them defensible, and until one exists a Profile that requires a tighter bound than 7.1 names is conforming and a Profile that requires a looser one MUST record the departure under G8.

7.0.3 · Failure. A run that cannot complete, or a set whose U/median exceeds the bound, produces **no reading**. A failed measurement enters the record as a BoundaryRecord under Section C with cause `instrument-failure`, and never as evidence that the field was below its floor.

7.0.4 · Cost. A measurement consumes the subject. Each procedure states whether it requires exclusivity. Whether a measurement window counts against the availability obligation is a term of the agreement and MUST be fixed under A4.

7.0.5 · Filing. A reading is filed as an evidence receipt naming the procedure identifier, the conditions under 7.0.1, the n runs, the median, the scaled median absolute deviation, the Grade Class field it is offered against, and, where the procedure names its bound as U/median, the coverage factor k and the expanded uncertainty U.

7.0.6 · Traceability. Metrological traceability is the property of a result that relates it to a reference through a documented unbroken chain of calibrations, each contributing to the uncertainty [VIM]. **The procedures in 7.1 that measure a rate — MP-01 through MP-05 — have no such reference.** Their instrument is the subject under test or another node beside it, and no calibrated standard stands behind either. They establish **repeatability**: agreement of a reading with itself, under the conditions 7.0.1 requires it to declare.

A reading filed under MP-01 is evidence that a measurement was taken, was taken under declared conditions, and was repeatable within a stated bound. **It is not evidence that the quantity measured is what a calibrated instrument would have read**, because no such instrument exists for these quantities today.

7.0.7 · Commencement and cadence. A reading taken at commencement under B8.5 and a reading taken at cadence answer different questions, and the Profile MAY hold them to different bounds, subject to 7.0.2.3.

Legal metrology keeps this distinction: an instrument entering service is held to a tighter tolerance than the same instrument in service, so that a limited amount of drift is permitted before the instrument is rejected. **The commencement reading establishes that each V1 field of the Grade Class contracted for was delivered, and it is the baseline every later reading is read against.** Where the Profile states one bound for both, it MUST say so.

7.1 · Procedures.

ID	Establishes	Probe	Runs	Bound; for MP-01 to MP-05, U/median at $k = 2$	Exclusivity
MP-01	`memory_bandwidth`	Sustained read and read-write sweeps over a working set exceeding last-level cache by ≥ 8×	≥ 5	2 %	Per subject
MP-02	`link_bandwidth_per_gpu`, `coherent_domain_size`	Pairwise peer-to-peer bandwidth matrix across all accelerators on the subject	≥ 3 full matrices	3 % per ordered pair	Per subject
MP-03	`fabric_bandwidth_per_gpu`	Pairwise inter-node transfer across the allocation, all-to-all and ring	≥ 3	5 %	Across ≥ 2 subjects held simultaneously
MP-04	`storage_throughput_floor`, `storage_iops_floor`, `local_nvme_per_gpu` throughput	Sequential and random access at a queue depth named in the Filing Plan, working set exceeding any cache tier	≥ 5	5 %	Per subject
MP-05	`egress_floor`, `ingress_floor`	Sustained transfer against an endpoint outside the provider's network, named in the Filing Plan	≥ 5	10 %	None
MP-06	accelerator identity, `die_topology` class	Per-streaming-multiprocessor latency map and cache-bypassing memory sweep, per Annex 6	≥ 3	Identity: match to the 7.1.5 baseline. Class: unanimous across runs	Per subject
MP-07	coarse geographic position	Round-trip timing against public network landmarks, with decoy sites named in advance	≥ 1 landmark set	Landmark set stated	None

7.1.1 `coherent_domain_size` is read from MP-02 as the largest set of accelerators mutually reachable at the link bandwidth of A8.2, rather than at host-bus bandwidth. **A domain is what the instrument finds, not what the control plane reports.**

7.1.2 MP-03 requires the customer to hold two subjects at once. A customer holding one subject cannot establish fabric bandwidth, and for the duration the field reaches that customer's record as a V4 field does, by a signed statement under A8.11.

7.1.3 MP-07 is admissible only where the landmark set and the decoy set were **named before the reading was taken**. A location claim tested only against landmarks chosen afterwards establishes nothing, since the choice can follow the answer.

7.1.4 A procedure establishes that a field met its floor. **It does not establish that the field held between readings.** Continuity is a matter for cadence under B8.5.

7.1.5 · MP-06 baseline. An identity reading is a comparison, so it requires something to compare against. The baseline is the per-subject fingerprint taken at commencement under B8.5 and committed under the A10 digest before the Window opens. **An identity reading taken against a baseline committed after the Window opened establishes nothing**, since a baseline recorded later can follow the substitution it would otherwise reveal. The `die_topology` class is established where three runs return one class and no run returns another.

7.1.6 · MP-07 consensus. A location reading is admissible only where the fit discards probes whose reported position is inconsistent with their own round-trip time, and the decoy set is scored on the probes that survive. In the source's experiment, round-trip time taken alone left two of eleven decoys standing (Annex 6).

7.2 · What the procedures do not reach. `allocation_contiguity`, `tenancy`, `preemptibility` and rack-level position have no procedure here. They are V4 under A8.10 and carry a required evidence class under A8.11.

MP-02 finds which accelerators are mutually reachable at link bandwidth. **It does not find where they are.** MP-07 places a subject at a coarse geographic position. **It does not place it in a rack.** No procedure in this annex reports another tenancy.

This annex is the boundary of what measurement does for a Grade Class field under this standard. Everything beyond it is a signature or nothing.

Annex 8 · Profile conformance checklist

Every term this standard requires a Contract Profile to fix, with the clause that requires it and the set it is drawn from. **A Profile missing any row marked required cannot be armed**, and the omission is detectable when it is armed rather than when it is disputed.

Rows 25 to 29 are required by this annex itself, and for them the Clause column names the section each term attaches to. Every other row restates a requirement of the clause it cites.

8.1 · Required — the Profile cannot be armed without these.

#	Term	Clause	Drawn from
1	Obligation reference	A1	the agreement
2	Period function	A2	metric registry, `canonical` or declared `experimental`
3	Measurement window, offset, edge treatment	A3	the agreement
4	Exclusions, each with the evidence required to invoke it	A4	the agreement
5	Tolerance and threshold	A5	the agreement
6	**Materiality threshold** — form and its parameters	A6	**Annex 2.2**
7	**Unresolved ceiling** — form and its parameters	A6	**Annex 2.3**
8	Remedy schedule reference	A7	held where the evaluation cannot read it
9	**Grade Class** — every field valued or marked `not-material`	A8	Annex 4, or declared
10	**Required evidence class for every V4 field**	**A8.11**	the agreement
11	Substitution factor, where any is permitted	A9.2	**Annex 2.6**
12	**Standard digest, committed before the Window**	**A10**	computed over A1–A9

#	Term	Clause	Drawn from
13	Measured Subject grain	B3	the agreement
14	**Covered Layers**	B7.1	L1–L7; L8 is never covered
15	Minimum customer-side observation	B8.2	the agreement, or B8.2's default
16	V1 reading cadence	B8.5	the agreement
17	**What expiry costs the silent party**	**I.1.1**	**Annex 2.4**
18	**Evidence-fault allocation, per obligation**	**J1**	**Annex 2.5**
19	**Divergence resolution** — form and its parameters	**H4**	**Annex 2.7**
20	**Conformity decision near a limit**, per measured field and for the A5 threshold where the Period Figure is built from measured values	**7.0.2.1**	**Annex 2.8**
21	**Reproducibility limit**, where both parties read one V1 field	**7.0.2.2**	the agreement
22	Attestation topology, per V2 field	6.0.2	passport or background-check
23	Review deadline	I.1	the agreement, falling after the delivery Claim is filed and before the correction window closes; or I.1's W+12
24	Recomputation passes, where more than one	I.1.4	the agreement; one pass where unstated
25	Settlement policy quorum	F2	order-insensitive forms only
26	Challenge policy and post-finality window	K	the agreement
27	Grant constitution	F3	absence is refusal, not permission
28	Head publication frequency and gossip obligation	E2	at minimum, one head per Window close
29	Value at risk for the obligation	F1	the agreement
30	Whether a measurement window counts against availability	7.0.4	fixed under A4
31	Retention period	K1	not less than this standard's floor
32	Declared departures from this standard	G8	**"none" is a permitted entry**
33	**Materiality form and period function in a combination Annex 2.2.1 permits**	**2.2.1**	MT-01 or MT-05 with any function; MT-02 where every Filing Plan in force for each Window samples at the A3 interval, G10 refusing a modification that would change it; MT-03 or MT-04 with F-05 or F-06
34	**One evidence-fault form per obligation**	**J1**	**Annex 2.5**, one form
35	Credit Claim review span	I.1.5	the agreement, at least one business day; or the Window's own span under I.1.5
36	**Materiality reference party**, under MT-01 to MT-04	**A6**	the provider or the customer, named under the A10 digest

8.2 · The seven that decide who pays. Rows 6, 7, 10, 17, 18, 19 and 20 each settle who bears a cost when the record is partial, silent, contested, or measured close to a limit. **The standard requires of those seven that they exist, that they are fixed before the Window opens, and that all except row 10 are selected from a published set.**

Row 18 additionally carries a merits test. Under 2.5.1 a selection is refused where the party holding the required evidence is the party that gains from its absence. It refuses only the case where a party is paid to stay blind. Rows 33 and 34 limit rows 6 and 18 to the combinations under which a record yields one result, and row 36 names the figure row 6 is read against.

8.3 · Detection. Rows 1–36 are checkable without reference to any evidence or any Window, and every row but 33 against the Profile alone; row 33 also reads the Filing Plans in force, and G10 applies it to each later modification. **A Profile defect is therefore an arming-time**

finding, not a dispute-time discovery, and a compiler that arms Profiles is the right place to enforce this annex.

The alternative is the ordinary case. A term nobody fixed becomes a term somebody argues, six weeks after the month closed, on a call.

Annex 9 · Conformance vectors

Section 4 states what conformance means. The vectors this annex lists check it.

Published with this standard, under the MIT licence of 0.2:

File	Contents
`annex3-worked-month.json`	Annex 3's month as a fixture — the standard in force, the five events, and every figure the record produces. A conforming implementation reproduces all of them from the events and the standard terms the fixture carries, read with the inputs Annex 3 states.
`edge-cases.json`	Thirty-one cases stating an input and the only conforming outcome, each naming the clause it tests and what a conforming implementation MUST NOT return.
`MANIFEST.sha256`	A digest per file.

9.1 A vector states a required outcome and carries no authority beyond the clause it names. Where a vector and the clause it tests disagree, **the clause governs and the disagreement is an erratum under L6**.

9.2 Passing these vectors makes an implementation consistent with the cases they cover. **It establishes nothing about a case they do not cover**, and this set covers the arithmetic of one worked month and thirty-one boundary conditions.

9.3 No party certifies an implementation under this standard, and PSYCHE operates no conformance programme. **A claim of conformance is the claimant's own**, made against a named version, and a counterparty can run these vectors against it without asking anyone's permission.

References

Three kinds of entry: two relied on, and one named, not relied on. **Normative** references are those a conforming Profile or record depends on. **Informative** references support an argument this standard makes and bind nothing.

A reader who cannot obtain a reference cannot check the clause resting on it, so a freely available source is named wherever one carries the same substance. Where a reference is available only at cost and no free equivalent could be found, this standard says so rather than name a substitute that says something else.

Normative.

Tag	Reference
[**19086-2**]	ISO/IEC 19086-2:2018, *Part 2: Metric model*, with Amendments 1:2023 and 2:2025. Free from ISO. Clause 8 is the basis of Annex 2.1.
[**22123-1**]	ISO/IEC 22123-1:2023, *Cloud computing — Part 1: Vocabulary*. Free from ISO. Supersedes ISO/IEC 17788:2014. The source of *cloud service provider* and *cloud service customer*, which this standard calls the provider and the customer.
[**ATC**]	F. Alpay and T. Alpay, *Unprivileged Topology Certificates for Cloud GPU Attestation*, arXiv:2606.24934v1 [cs.CR], submitted 2026-06-22. Preprint, free from arXiv; this standard cites version 1 and no later one. The source of the primitive MP-06 and MP-07 rest on, set out in Annex 6. Cited in 0.3, where its status as third-party work is disclosed.
[**RFC6962**]	IETF RFC 6962, *Certificate Transparency*, §2.1. Free from the IETF. The Merkle algorithm, leaf and node hashing and audit path, that E2 runs over the reference implementation's own event and head formats, as Annex 5 states.
[**RFC9162**]	IETF RFC 9162, *Certificate Transparency Version 2.0*, §2.1.3.2. Free from the IETF. The verification of an inclusion proof under E2, run over the same formats, as Annex 5 states.

Informative.

Tag	Reference
[**19086-1**]	ISO/IEC 19086-1:2016, *Cloud computing — Service level agreement (SLA) framework — Part 1: Overview and concepts*, with Amendment 1:2025. Free from ISO. Cited in Annex 2.1.1 for the framework 19086-2's metric model belongs to.
[**HVR**]	The Hague Rules as amended by the Brussels Protocol 1968 (the Hague-Visby Rules), Article III. Public international convention; the text is scheduled to national carriage-of-goods statutes and is freely available. Cited in Section 2.
[**UCP600**]	ICC Uniform Customs and Practice for Documentary Credits, 2007 revision, Articles 14(a) and 34. Published by the ICC at cost. *Abani Trading Pte Ltd v BNP Paribas* [2014] SGHC 111, free from the Singapore judiciary, reproduces Article 14(a) at [22] and quotes the part of Article 34 cited here at [23].
[**CSCC**]	Cloud Standards Customer Council / OMG Cloud Working Group, *Cloud Service Agreements: What to Expect and What to Negotiate*, Version 3.0, 2019, p. 20. Free from omg.org. Cited in Section 2 for current practice on service-credit claims.
[**BJPD**]	*Brown Jenkinson & Co Ltd v Percy Dalton (London) Ltd* [1957] 2 QB 621 (CA). Cited in Section 2 for the effect of a knowingly false clean bill issued against a letter of indemnity.
[**GUM**]	JCGM 100:2008, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement*. Free from the BIPM.
[**VIM**]	JCGM 200:2012, *International vocabulary of metrology*. Free from the BIPM. The source of *metrological traceability* as used in Annex 7.
[**JCGM106**]	JCGM 106:2012, *The role of measurement uncertainty in conformity assessment*. Free from the BIPM.
[**RATS**]	IETF RFC 9334, *Remote ATtestation procedureS (RATS) Architecture*. Free from the IETF. The source of the attestation roles named in Annex 6.
[**DA**]	T. Laor et al., *DrawnApart: A Device Identification Technique based on Remote GPU Fingerprinting*, NDSS 2022. Cited in Annex 6 as independent precedent for the measurement technique.
[**CORES**]	T. Ristenpart, E. Tromer, H. Shacham and S. Savage, *Hey, You, Get Off of My Cloud! Exploring Information Leakage in Third-Party Compute Clouds*, Proceedings of ACM CCS 2009, pp. 199–212. Free from the authors. Cited in A8.11 as an existence result for co-residency detection.
[**GPUBOX**]	S. B. Dutta, H. Naghibijouybari, A. Gupta, N. Abu-Ghazaleh, A. Marquez and K. J. Barker, *Spy in the GPU-box: Covert and Side Channel Attacks on Multi-GPU System*, ISCA 2023, doi:10.1145/3579371.3589080. Available as an author copy at par.nsf.gov/servlets/purl/10428033. Cited in A8.11 as an existence result for contention across a shared interconnect.
[**MISO**]	Midcontinent Independent System Operator, *Market Settlements Business Practices Manual*, BPM-005, revision 29, effective 12 July 2026, §2.1.3. Free from MISO. Cited in I.1.4.2 for an operating day settled at 7, 14, 55 and 105 days.
[**CAISO**]	California ISO, *Fifth Replacement FERC Electric Tariff*, Section 11, *Settlements and Billing*, as of 5 February 2025, §§11.29.7.1 and 11.29.8.2(b). Free from CAISO. Cited in I.1.4.2 for the Initial Settlement Statement T+9B and the Recalculation Settlement Statements T+70B, T+11M, T+21M and T+24M.
[**AEMO**]	Australian Energy Market Operator, *Metrology Procedure: Part B — Metering Data Validation, Substitution and Estimation*, version 8.23, effective 1 November 2026, §2.4 (quality flag F), §2.5(b) and (h), and §4.3. Free from AEMO. Cited in Annex 2.5 for substitution by methods published in advance and for final substitute data with its exceptions.

Tag	Reference
[**GAFTA124**]	Gafta, *Sampling Rules No. 124: Rules for Sampling, Analysis Instructions and Certification*, effective for contracts dated from 1 July 2026, rule 9.5.3.4. Free from Gafta. Cited in Annex 2.7.3.
[**COMEX**]	CME Group, *COMEX Rulebook*, Chapter 7, *Delivery Facilities and Procedures*, Rule 707, *Obligations in Metal Delivery and Dispute Resolution*, paragraph (2). Free from CME Group. Cited in Annex 2.7.3.
[**ICE**]	ICE Futures U.S., Rules, Chapter 8, *Coffee "C" Rules*, Appendix II, *Procedures for Grading Coffee and Issuance of Certificates of Grade*, paragraphs (d), (i) and (m). Free from ICE. Cited in Annex 2.7.3.
[**FOSFA**]	FOSFA, *Rules of Arbitration and Appeal*, revised and effective from 1 April 2026, rule 2(e). Free from FOSFA. Cited in Annex 2.7.3.

Named, not relied on. **ISO/IEC 19086-3:2017**, *Part 3: Core conformance requirements*, is the part of that family nearest to Section 4 and Annex 8 of this standard. It is available only at cost, and no free source stating its requirements could be located as of 2026-09-03. **Section 4 was written without it**, and this standard makes no claim about the relationship between the two.

Send a closed case and PSYCHE will reconstruct it under SQCD-1.